Distribution, Technique, and the Transformation of Accumulation: Reconstructing Capacity Utilization in the Center and Periphery*

WORK IN PROGRESS / PLEASE DO NOT CIRCULATE

Diego Polanco†
Department of Economics, University of Massachusetts Amherst
dpolanconeco@umass.edu

August 10, 2026

Abstract

Capacity utilization has been treated either as a given macro ratio, an inherited benchmark, or an empirical residual. That leaves unresolved the problem of how accumulation is transformed into productive capacity under capitalist disequilibrium. This chapter reconstructs capacity utilization as a structurally mediated object, recovered through the identification of the transformation elasticity θ, where distribution mediates the conversion of accumulation into productive capacity. Applying this framework comparative-econometrically to the United States (1947–2024) and Chile (1920–2024), we demonstrate that the transformation elasticity is endogenously determined through the choice of technique and the balance of class forces. In the center, the shift from a wage-led Fordist nexus to a post-Fordist relative price channel led to institutional decoupling and overaccumulation. In the periphery, this transformation is structurally truncated by external balance constraints and balance of payments dynamics. Our findings challenge the view of utilization as a simple residual, showing it to be a key institutional marker of macro-political-economic reproduction.

Keywords: Capacity Utilization; Transformation Elasticity; Choice of Technique; Cointegration; Marx-Okishio; External Balance; Center-Periphery.
JEL Codes: B51; C32; E01; E22; E32; P16.

*This paper benefited from AI-assisted editing for syntax refinement, structural suggestions, and stylistic consistency checks. All analytical arguments, historical interpretations, and substantive content remain the author's sole responsibility.

†Doctoral Candidate in Economics, Department of Economics, University of Massachusetts Amherst.

1 The Unresolved Problem of Capacity Utilization

A This paper examines how the rate of profit, capital accumulation, and productive capacity formation interact under conditions of distributional conflict between labor and capital.

B

[**PLACEHOLDER**] Section 1 is unwritten. What exists above is a single orienting sentence; the section that the locked outline requires does not yet exist. The remainder of this box states what it has to do, so the gap is legible rather than silent.

C

[**PLACEHOLDER**] **Burden of the section.** Establish that capacity utilization is unresolved in the treatments that have come down to us, in three distinct ways, and show that these are three faces of one problem rather than three separate difficulties. First, utilization treated as a *given macro ratio*: taken as observed when its denominator is not. Second, utilization treated as an *inherited benchmark*: anchored to an engineering or survey norm imported from outside the accounting framework that is supposed to give it meaning. Third, utilization treated as an *empirical residual*: whatever is left after output has been fitted to capital, which presupposes the very transformation relation at issue. The third case is where the filtering literature sits, and the objection to it here is structural rather than methodological — a filter that fits output to capital imposes a unit transformation elasticity by construction, and so conflates utilization with the object that would be needed to identify it. That argument is available from this chapter's own accounting framework and requires no external authority.

D

[**PLACEHOLDER**] **Bottom line up front.** Per the writing calibration in force, the contribution must be visible by the end of the second paragraph: what the paper examines, how it identifies the transformation elasticity, and what the comparison of the United States and Chile establishes. The reader should not have to reach Section 3 to learn what is being claimed.

E

[**PLACEHOLDER**] **What must not appear here.** Long literature catalogues. The full center–periphery discussion, which belongs in Section 5. Estimator debates, which belong in Section 4 and Appendix C. Detailed methodological caveats. The section motivates a problem; it does not litigate the solution.

F

[**PLACEHOLDER**] **Closing-line burden (locked).** The section must arrive at: *the utilization problem cannot be resolved at the level of the ratio alone; it must be reconstructed through the relation that governs productive-capacity formation.* This hands off directly to Section 2.

G

[**PLACEHOLDER**] **Also unresolved: the abstract's sample window.** The abstract states a United States window of 1947–2024, while Sections 4 through 6 estimate over 1929–2024 with MATHPLACEHOLDER0XYZ. The two cannot both be right. Note further that 1929–2024 spans 96 years against a reported MATHPLACEHOLDER1XYZ of 94, and that the estimation prose elsewhere gives the effective span as 1931–2024 after lag-1 conditioning and listwise deletion — which is consistent with MATHPLACEHOLDER2XYZ and suggests the table titles, not the sample, are what needs correcting.

2 The Internally Divided Firm and Capacity Formation

2.1 Capacity utilization is not a technical parameter. It is an inter-scalar political-economic settlement. The macroeconomic capacity ceiling—the denominator of the utilization rate—is not a neutral engineering constraint but a historically contingent outcome conditioned by the balance of class power. Yet identifying this distributional conditioning at the macro level leaves the underlying transmission mechanism black-boxed: how, precisely, does distributive conflict physically alter the conversion of accumulated capital into productive capacity?

2.2 Answering this question requires dismantling a hidden premise shared by virtually all heterodox growth models: the unified firm. Whether treating the normal rate of capacity utilization as an endogenous behavioral target or an exogenous gravitational anchor, standard frameworks presume a coherent organization capable of selecting a utilization benchmark, enforcing it across the labor process, and sustaining it over time. Formally, the utilization rate is defined as:

$$\mu_t \equiv \frac{Y_t}{Y_t^p}, \tag{1}$$

2.3 where Y_t is realized output and Y_t^p is potential productive capacity. But this ratio conceals a deeper question: who determines Y_t^p, and under what organizational conditions?

2.4 The unified-firm premise collapses when confronted with the social organization of capitalist production. In a decentralized economy, individual capitals commit fixed investments without *ex-ante* coordination, confronting one another in competitive markets. Within the enterprise, production decisions encounter contradictory managerial and worker imperatives. Capacity utilization cannot be reduced to a target selected by a single optimizing entity. To theorize the political economy of productive capacity, we replace the unified firm with two interconnected frameworks: Okishio's theory of decentralized market anarchy, and Vidal's theory of the internally divided firm.

2.1 Smooth Reproduction and the Unified-Firm Premise

2.5 Proportional reproduction is a condition of feasibility, not a tendency of capitalist accumulation. Smooth reproduction requires strict material proportionality between departments of production—a condition Basu (2022) and Okishio (2022) formally established defined by the latter as the equilibrium accumulation trajectory. Bergmann et al. (2009) call this balanced trajectory, where realized output meets productive capacity ($\mu_t = 1$), the *"golden path."* But the golden path is a methodological closure, not a gravitational attractor. It identifies the conditions under which realization can reproduce itself smoothly; it does not describe the path capitalist accumulation actually follows. Neither the post-Keynesian assumption of long-run balanced growth (Blecker and Setterfield, 2019) nor the *"Keynesian in the short run, classical in the long run"* synthesis (Duménil and Lévy, 1999) captures this persistence of unbalanced expansion.

2.6 Nothing in the logic of decentralized competition establishes that private investment decisions will reproduce these proportions. A stylized illustration makes the point concrete. If productive capacity grows at 5% per year while aggregate demand for output grows at 2%, capacity outpaces demand by three percentage points each year. After a decade, utilization falls to roughly 75% of

capacity—not because of a temporary shock, but because uncoordinated investment persistently builds capacity faster than demand can absorb it. Balanced growth ($g_Y = g_K$) is merely one configuration of the transformation elasticity θ; it is not a restoring force. Smooth reproduction ($\mu_t = 1$) does not require $\theta = 1$; it holds for any transformation elasticity. Balanced growth is the narrower special case in which $\theta = 1$ and $\mu_t = 1$ coincide in the long run.

2.7 Because private investment decisions are made without *ex-ante* coordination, real capitalist economies routinely violate the proportional conditions of the golden path. Okishio (2022) demonstrates that chronic disproportionality is not a temporary friction but the structural norm of decentralized accumulation. Treating the transient benchmark of $\mu_t = 1$ as a behavioral target therefore relies on a hidden premise: that the firm is a unified planning subject capable of selecting a utilization norm, enforcing it across the labor process, and sustaining it over time. This premise—the unified firm—is the target of the present section.

2.8 The denominator of the utilization ratio (productive capacity Y_t^p) is not an engineering constant. It is a contested settlement of divided management and shop-floor struggle. The numerator (realized output Y_t) is hostage to the anarchy of demand. Smooth reproduction supplies a reference case against which this historical non-coincidence can be studied. It does not supply the mechanism that produces it.

2.2 Decentralized Accumulation and Market Anarchy

2.9 If the "golden path" of smooth reproduction is merely a benchmark of feasibility, the actual trajectory of capitalist accumulation is governed by the decentralized organization of investment. Capitalist production is coordinated through exchange, meaning that the social validation of private decisions occurs only *ex-post*. Because money mediates exchange, sale and purchase are separated moments of the capital circuit (Basu, 2022). Surplus value must be realized in circulation, not by construction (Foley, 1985; Trigg, 2020). This constitutes the logic of capital as impersonal domination: capitalists commit to production and investment with the law of value operating *"behind their backs."*[1] They are subject to market forces and value relations they neither control nor consciously direct. Critically, this anarchy is not a coordination failure in an otherwise functional market. The market is a socially constituted institution whose normal operation produces disproportionality as a structural feature (Vidal and Peck, 2012). Okishio (2022) locates the source of macroeconomic instability precisely in this constitutive anarchy. Individual capitals commit to fixed investment on the basis of private profititability expectations, yet the market validates those commitments only after production has begun. No firm can know in advance whether its expansion fits the investment and consumption decisions taken elsewhere. The coordination problem is not one of imperfect information surrounding an otherwise coherent plan; no economy-wide plan exists.

2.10 Smooth reproduction requires proportional expansion across sectors producing means of production and sectors producing consumption goods. Because private investment does not enforce this proportionality, firms extrapolate past prices, profit rates, and sales. A signal that induces expansion in one branch routinely produces excess capacity before complementary demand materializes

[1] Karl Marx, *Capital*, Volume I, Chapter 1, Section 4 (*"The Fetishism of Commodities and the Secret thereof"*), where the law of value operates as an overriding law of nature when private producers exchange products (Marx, 1999).

elsewhere. Contraction works through the same cumulative mechanism: falling sales cut investment, reducing aggregate demand and validating further cuts. Removing Say's Law reintroduces realization as a substantive condition of reproduction: accumulation may generate capacity that demand fails to absorb, or demand may temporarily exceed productive capacity (Foley, 1985). Prices register the resulting imbalance, but they do not prevent the sunk commitments produced by it.

2.11 Fixed capital gives this temporal lag its structural force. Structures and machinery embody long-horizon commitments that cannot be reversed when relative prices or demand conditions shift. Once installed, excess capacity leaves the firm to operate below its anticipated scale, defer maintenance, hold equipment idle, or devalue the asset. These responses distribute the costs of market anarchy among firms, workers, and creditors, but they do not restore the missing *ex-ante* coordination.

2.12 Idle capacity is therefore a recurrent, structural form of decentralized accumulation rather than a temporary friction. Private investment physically builds capacity before the market can realize its social use, recurrently a gap between installed productive capacities (Y_t^p) and effective demand (Y_t). The capacity utilization rate (μ_t) registers this gap at the macro level. Unbalanced growth is not a disturbance around proportionality; it is the structural consequence of distributively conditioned investment under decentralized decision-making. The analytical problem is therefore not how proportionality re-emerges, but how persistent imbalance remains historically reproducible. The condition $\mu_t = 1$ is at most a transient point through a given trajectory proportionality, while persistent divergence records the ongoing reality of market anarchy.

2.13 Yet Okishio's macro framework leaves a crucial gap: it does not explain how firms translate this external anarchy into internal operating decisions. The firm does not manage market anarchy; it internalizes it. A firm does not receive a price signal and respond as a frictionless, unified unit. It processes the signal through an organization divided over control, cooperation, effort, and the allocation of resources. Vidal's internally divided firm provides that missing mediation.

2.3 The Internally Divided Firm

2.14 Vidal builds on Marx's Volume I theory of the labor process, drawing on contemporary organizational theory and sociological institutionalism to develop an organizational political economy of the capitalist firm (Vidal and Peck, 2012; Vidal, 2022). Marx established that capitalist management faces a constitutive contradiction: it must simultaneously organize the cooperative labor required for production and enforce the coercive control required for valorization. Vidal (2022) demonstrates how this contradiction fractures the modern capitalist firm, treating it as a divided social organization whose internal structure is shaped by cognitive, normative, regulative, and organizational institutions. Management must extract surplus value through control over labor, but production also requires workers' cooperation, tacit knowledge, and problem-solving capacity. Standardization and surveillance secure a measurable pace of work while weakening the discretion needed to sustain complex production. The firm cannot maximize both imperatives at once because the labor process that produces value is also the site where workers contest its extraction. This contradiction is not a universal constant of capitalism; it is institutionally mediated, taking variegated forms across distinct accumulation regimes throughout historical capitalism (Vidal, 2019).

2.15 Workers respond to this contradiction by actively calibrating effort, cooperation, machine pacing, and the deployment of tacit knowledge. Capitalist management extracts potential capacity through concrete shop-floor mechanisms: increasing machine speed, deferring maintenance to hit short-term throughput targets, and enforcing rigid surveillance. Workers contest this extraction through labor turnover, collective exit, or the deliberate withdrawal of problem-solving cooperation and effort. Throughput, quality, and maintenance therefore depend on social relations that no engineering specification fixes in advance. A production target that raises machine speed by deferring maintenance may increase current output while degrading sustainable capacity. A directive that suppresses worker discretion may meet a short-term labor-control objective while worsening workflow bottlenecks. Labor effort is consequently not a fixed technical parameter applied to a pre-existing capital stock; it is an active, contested determinant of the capacity utilization itself.

2.16 This dynamic enforces a strict wedge between *engineering capacity* and *practical capacity*. Engineering capacity records the theoretical maximum output of equipment under ideal technical conditions. Practical capacity records the sustainable output rate dictated by actual maintenance schedules, product-mix complexity, and the negotiated intensity of labor. When managers report capacity to national statistical agencies, they rely on conventions shaped by their organizational position and the immediate pressures of the shop floor. Aggregate capacity statistics therefore do not reveal a neutral physical ceiling; they aggregate situated, institutional judgments (Shaikh, 2016; Nikiforos, 2016). Complementarily, empirical surveys of managerial practice confirm that firms do not systematically organize production decisions around a coherent capacity utilization target (Singh et al., 2022). For the purposes of macroeconomic modeling, the productive capacity ceiling (Y_t^p) is not an exogenous engineering constraint but an endogenous settlement of workplace conflict. This settlement is the micro-foundation through which class struggle and distribution transmit into capacity utilization, setting up the distributive–technological mapping developed in the following subsection.

2.17 Because the valorization–labor process contradiction cannot be solved, firms settle into organizational configurations of *"mediocre sufficiency"* (Vidal, 2022). These are hybrid arrangements that reproduce production without resolving the underlying conflict between control and cooperation. Partial worker autonomy coexists with rigid discipline; continuous improvement programs coexist with brutal throughput pressure; machinery might remain idle alongside binding productive bottlenecks. Following Vidal (2014), these arrangements are best understood as *institutional incoherence*: the firm's internal settlements are functional enough to sustain accumulation yet contradictory enough to generate persistent friction. This stands in direct contrast to the Varieties of Capitalism assumption of institutional complementarity, where national institutions are presumed to form coherent, mutually reinforcing systems (Hall and Soskice, 2003). Here, incoherence is not a deviation from an optimizing norm; it is the permanent, structural form through which capitalist management physically converts accumulation demand into capacity usage.

2.4 Distribution, Technique, and the Transformation of Accumulation

2.18 The choice of technique connects the divided firm to distribution, governing how accumulated capital is physically transformed into productive capacity. Capitalists adopt a new technique only when it reduces the cost of production at prevailing prices and wages; such a technique is capitalistically viable (Okishio, 1961). Okishio (1961) demonstrates that a viable capital-using,

labor-saving (CU-LS) technique raises the equilibrium profit rate when the real wage remains constant—a direct challenge to Marx's law of the tendential fall in the rate of profit. Marx's account, by contrast, allows the same form of technical change to depress profitability when rising capital intensity increases the capital advanced relative to surplus value. Basu (2022) formalizes the Marx–Okishio threshold: the rate of real-wage growth that separates these outcomes. It is a distributive boundary set by the balance of class forces, not a technical constant.

2.19 This boundary situates the present framework within the induced-innovation tradition while departing from it. The Innovation Possibility Frontier (IPF) literature, initiated by Kennedy (1964), treats distribution as selecting the *direction* of technical change along a given frontier: relative factor costs determine whether innovation is labor-augmenting or capital-augmenting, while the frontier itself—the scale of feasible productivity growth—remains exogenous (Julius, 2005; Tavani, 2012). Later assessments sharpen this limitation: IPF-based models explain the composition of technical change but remain largely silent on innovation intensity (Tavani and Zamparelli, 2018). Zamparelli (2024a) partially endogenizes scale by mapping a family of IPF levels to R&D effort, yet the analytical structure continues to rely on frontier geometry and steady-growth closure (Zamparelli, 2024b). An alternative strand reframes inducement through mechanization rather than frontier selection: Kaldor (1957) links productivity growth to capital deepening, while Cajas-Guijarro (2024) reformulates the IPF with an explicit choice over mechanization growth under distributive and labor-market conditions. The present framework builds upon a mechanization productive frontier, treating the distributive–technological relation as a regime-specific transformation between accumulation and productive capacity. Rather than selecting along a given frontier, distribution conditions mechanization intensity, which in turn determines the transformation elasticity linking accumulation to the formation of productive capacities.

2.20 The Marx–Okishio threshold gives wage pressure a precise role in mechanization, but the profitability outcome depends on the labor market closure. Basu (2026) demonstrates that the equilibrium profit rate after viable CU-LS technical change is governed by the institutional structure of capital–labor bargaining. Under a constant-wage-share closure—what Basu (2026) formalizes as the "strong labor" regime, applicable to Atlantic Fordism—any viable CU-LS technique unambiguously *lowers* the equilibrium profit rate. Marx's LTFRP reigns supreme precisely where labor is powerful enough to compel real wages to grow in tandem with productivity. Under this closure, rapid mechanization is induced by distributive pressure, yet the same pressure contests capital's claim on the resulting productivity gains (Vidal, 2015, 2019). A stylized illustration makes the mechanism concrete. If a 10% real wage increase raises mechanization intensity by 8% under strong-labor conditions, the transformation elasticity θ rises from 0.70 to 0.85: accumulation translates more efficiently into productive capacity because firms substitute machinery for labor. Wage-led mechanization was not a harmonious productivity bargain; it was a contested settlement in which the distributive pressure that induced technical change simultaneously eroded the profit rate it was meant to restore.

2.21 The weakening of organized labor after the Fordist crisis altered that mechanism. Under the "weak labor" closure—where the unemployment rate remains constant and workers lack the institutional power to enforce a target wage share—the distributive pressure to replace labor is suppressed (Basu, 2026). Yet mechanization does not cease. Lower relative machinery prices and

technical change in capital-goods production continue to favor capital-using techniques through a second, independent channel. Distribution and machinery prices therefore constitute a dual-channel mechanism in the choice of technique. Their relative force changes across accumulation regimes, which prevents the wage share from being treated as an external control added to an otherwise autonomous technical relation. Extending the stylized example: under Post-Fordist conditions, a 15% decline in relative machinery prices raises mechanization intensity by 6%, partially compensating for the weakened distributive channel and sustaining θ near 0.75 despite wage suppression. The same 10% wage increase that produced an 8% mechanization response under Fordism produces only a 2% response under Post-Fordism, leaving θ near 0.72. The two channels do not simply add; they interact through the institutional configuration that governs their relative force.

2.22 These channels still operate through the contradictions identified in the preceding subsections. Firms choose techniques before demand validates the associated capacity, and they operate those techniques through labor processes that divide control from productive cooperation. Investment consequently does not translate one-for-one into a stable productive ceiling. The transformation depends on the composition of capital, the distributive conditions governing mechanization, and the organization through which installed equipment becomes usable capacity. In a labor surplus economy—the third closure identified by Basu (2026), applicable to peripheral capitalism—the real wage remains constant and the equilibrium profit rate rises unambiguously after viable CU-LS technical change. Yet the peripheral transformation is structurally truncated by external financial subordination and the balance of payments constraint, a mechanism developed in the following subsection.

2.23 Okishio and Vidal jointly rule out any direct, automatic passage from accumulation to productive capacity. Decentralized investment generates structural disproportionality across firms and sectors; the internally divided firm internalizes this disproportionality as contested operating conditions rather than a coherent utilization target. The analytical object that captures this mediated transformation is the *transformation elasticity* ($\theta \equiv dy^p/dk$), which governs how accumulation converts into productive capacity under distributive conflict and market anarchy. The empirical analysis bounds this multi-sector mechanism to a single-sector aggregate, masking internal sectoral divergences while preserving the core transformation logic. Section 3 formalizes θ within a regime-specific analytical framework and derives its econometric implications through structural change controls.

2.5 Historical Regimes of Institutional Mediation and the Transformation Elasticity

2.24 The transformation elasticity is not a fixed engineering parameter. It is an emergent property of class struggle and the prevailing institutional environment. Decentralized accumulation does not automatically guarantee proportional reproduction ($\theta = 1$). Institutional mediation—wage-setting norms, financial architectures, and competitive structures—conditions how distributive pressures translate into mechanization choices (Aglietta, 2000; Vidal and Peck, 2012). Shifts in this mediation across historical periods will not be imposed through ex ante sample splits. The empirical strategy identifies them endogenously through structural change controls and break dummies, allowing the data to reveal how institutions bounded structural drift without requiring a formal equilibrium closure.

2.25 Basu (2026) demonstrates that the equilibrium profit rate after viable capital-using, labor-saving (CU-LS) technical change is governed by the labor market closure, which maps directly onto distinct historical accumulation regimes. Under a *constant-wage-share* closure—the institutional signature of Atlantic Fordism (1947–1973)—workers possess the bargaining power to grasp productivity gains, forcing real wages to grow in tandem with labor productivity. In this strong-labor regime, viable mechanization unambiguously lowers the equilibrium profit rate, so that Marx's law of the tendential fall in the rate of profit holds (Basu, 2026). This stagnation tendency was institutionally offset: the class compromise, mass consumption, oligopolistic market structures, and military Keynesianism of the Fordist era absorbed the surplus and sustained effective demand (Vidal, 2019).

2.26 Under a *constant-unemployment* closure—the institutional signature of the post-Fordist, neoliberal era—the class compromise is dismantled and labor's bargaining power is severely weakened. The wage share falls, and mechanization is driven not by distributive pressure but by the relative cheapening of ICT and the shift toward financially reversible, short-horizon assets (Lapavitsas, 2011; Pagano, 2014). Because wages no longer track productivity, the profit rate may stabilize or rise, but the suppression of wage-led demand generates a chronic under-consumptionist stagnation tendency. This regime produces a dysfunctional utilization configuration: accumulation decouples from productive capacity formation, generating chronic idle capacity that is temporarily masked by debt-led consumption and financial fragility rather than sustainable realization (Stockhammer, 2012; Vidal, 2019).

2.27 In the capitalist periphery, the relevant framework is the *constant-real-wage* closure, which corresponds to Okishio's original assumption and maps onto a surplus-labor economy (Lewis, 1954; Basu, 2026). During the import-substituting industrialization (ISI) era, the *rural-urban* transition regulated the peripheral transformation, providing an unlimited supply of labor from the countryside that disciplined urban wages and allowed the profit rate to rise alongside mechanization. However, peripheral mechanization is structurally import-biased: widening the mechanization gap requires imported capital goods and technologically non-substitutable inputs (Prebisch, 1981). The balance of payments constraint therefore acts as a hard external ceiling on the transformation of accumulation into capacity. Under a developmentalist regime (1933–1970), this operated as an *import capacity ceiling*; under neoliberal subordination (1974–present), the WTO and bilateral free trade agreements legally encoded a *policy space ceiling*, locking the periphery into extractive specialization and premature deindustrialization (Ahumada, 2019; Palma, 2019).

2.28 These historical configurations establish that the productive ceiling is not a neutral, trans-historical engineering constraint but a contingent outcome of decentralized accumulation, class power, and a country's relative positionality within the capitalist world system. The center–periphery comparison reveals a fundamental asymmetry: in the center, structural crises arise from the *exhaustion* of institutional mediation (e.g., the 1970s profit squeeze); in the periphery, crises arise from the *absence* of sovereign mediation and permanent truncation by global financial markets. The immediate task is to translate these historical-institutional claims into a formal mathematical architecture. Section 3 formalizes induced technological change under a cost-minimization problem, first advanced by Cajas-Guijarro (2024), in three stages of increasing complexity: i) aggregate capital-deepening; ii) mechanization bias under heterogeneous capital decision-making; and iii) a

peripheral foreign-exchange bottleneck that induces technological change through mechanization. These analytical foundations provide the toolkit to deploy an empirical strategy identifying a state-dependent θ parameter endogenous to distributive conflict, tracing the trajectory of productive capacities and the capacity utilization ratio through a structural empirical approach.

3 The Analytical Framework: Formalizing the Transformation Elasticity

3.0 This section translates the distributive–technological mechanism of Section 2 into a formal model that yields estimable restrictions. Cajas-Guijarro (2024) first advanced the cost-minimization problem over mechanization growth; we extend it through three subsections of increasing complexity. Section 3.1 formalizes aggregate capital-deepening under Leontief technology and derives the transformation elasticity θ endogenously from the cost-minimizing choice of technique. Section 3.2 unbundles aggregate capital into extensive nonresidential structures and intensive machinery, and derives the state-dependent mechanization gap κ_t under a dual-channel viability condition. Section 3.3 imposes the peripheral foreign-exchange bottleneck as a binding constraint on κ_t, generating a corner solution that truncates θ. Each subsection nests the preceding one: setting the heterogeneity parameter to zero recovers the baseline model; relaxing the foreign-exchange constraint recovers the center model. The empirical analysis bounds this multi-sector mechanism to a single-sector aggregate, which masks internal sectoral divergences but preserves the core transformation logic. Section 3.4 synthesizes the theoretical claims of the three models. Section 4 consolidates the identification strategy and derives the cointegrating specifications.

3.1 The Baseline Model of Cost-Minimizing Mechanization (Homogeneous Capital Stock)

3.1 Actual aggregate output is represented by a Leontief production function with fixed technical coefficients:

$$Y = \min\left(AL,\ \mu BK\right) \tag{2}$$

where $A = Y/L$ is actual labor productivity, $\mu = Y/Y^p$ is capacity utilization, and $B = Y^p/K$ is maximum capital productivity. This production structure governs the rate of profit through the fundamental heterodox accounting identity:

$$r = \mu B(1 - \omega) \tag{3}$$

where ω is the wage share. Standard macroeconomic models treat B as an exogenous technological given and μ as a behavioral constant smoothly targeted by a unified firm. We reject both assumptions. The components of (3) are endogenous outcomes of class struggle and decentralized market anarchy, as established in Section 2. This subsection formalizes how distributive pressure forces cost-minimizing firms to alter their choice of technique, endogenously determining B.

3.1.1 Tactical Choice of Technique and Induced Innovation

3.2 Firms select a mechanization growth rate q to maximize the net productivity surplus. Let a denote the growth rate of labor productivity. The effective distributive cost of deploying mechanization

growth is $q(1-\omega)$. The firm solves:

$$\max_{q} \; c(q) = a - q(1-\omega) \tag{4}$$

This objective captures the induced-innovation logic: firms accelerate mechanization to expand labor productivity growth, but the distributive cost of capital intensity holds them back. A technological frontier bounds this choice:

$$a = \Phi(q) = \lambda_0 + \lambda_1 q + \lambda_2 q^2 \tag{5}$$

3.3 Assumption 1. *Concavity of the mechanization frontier.* The frontier parameters satisfy $\lambda_0 > 0$, $\lambda_1 > 0$, and $\lambda_2 < 0$. The base productivity growth λ_0 sets the floor without advanced machinery. The linear coefficient λ_1 captures the direct productivity payoff of mechanization. The strict concavity $\lambda_2 < 0$ ensures diminishing marginal returns to mechanization.

3.4 Substituting (5) into (4) and solving the first-order condition yields the optimal mechanization growth:

$$c(q) = \lambda_0 + [\lambda_1 - (1-\omega)]\, q + \lambda_2 q^2 \tag{6}$$

$$\frac{dc}{dq} = \lambda_1 - (1-\omega) + 2\lambda_2 q^* = 0 \tag{7}$$

$$q^*(\omega) = \frac{(1-\omega) - \lambda_1}{2\lambda_2} \tag{8}$$

The strict concavity ($c''(q) = 2\lambda_2 < 0$) guarantees a unique maximum.

3.5 Proposition 1. *Induced innovation.* Under Assumption 1, the optimal mechanization growth is strictly increasing in the wage share:

$$\frac{\partial q^*}{\partial \omega} = -\frac{1}{2\lambda_2} > 0 \tag{9}$$

A rise in the wage share forces cost-minimizing firms to substitute machinery for labor. Rising wages drive mechanization directly through the first-order condition, pushing optimal capital intensity growth upward. *Proof.* Immediate from differentiating (8). ■

3.X Remark 1. The objective (4) does not assert that the firm is a coherent optimizer. Following Vidal (2022), the firm is internally divided between the *control imperative* (valorization: cost reduction through mechanization) and the *cooperation imperative* (production: the need for worker effort, tacit knowledge, and problem-solving capacity). The formal model captures the cost-minimizing pole—the Okishio viability logic—while the constraints on the optimization problem capture the limits imposed by the cooperation imperative and by external conditions. The firm maximizes net productivity surplus—labor productivity growth minus the distributive cost of mechanization—subject to diminishing returns. The firm's actual mechanization choice is a contested settlement between these imperatives, not a coherent optimum. What the model demonstrates is the cost-reducing direction in which distributive pressure pushes the settlement, not the settlement itself.

3.1.2 Endogenous Transformation Elasticity

3.6 Substituting the optimal technique back into the technological frontier recovers the normal growth rate of labor productivity under smooth reproduction:

$$a^N(\omega) = \lambda_0 + \lambda_1 q^*(\omega) + \lambda_2 \left[q^*(\omega)\right]^2 \tag{10}$$

This optimal technique dictates the structural trajectory of the Leontief parameter B. By definition, maximum capital productivity is $B = Y^p/K$. In dynamic growth rates, the growth of capital productivity is:

$$b \equiv g_{Y^p} - g_K = a - q \tag{11}$$

Evaluated at the smooth reproduction benchmark: $b^N(\omega) = a^N(\omega) - q^*(\omega)$.

3.7 Definition 1. *Transformation elasticity.* The transformation elasticity θ is the structural parameter that translates capital accumulation into productive capacity growth:

$$g_{Y^p} = \theta\, g_K \tag{12}$$

Substituting (12) into (11) reveals the link between the level coefficient B and the elasticity θ:

$$\theta = 1 + \frac{b}{g_K} \tag{13}$$

3.8 To evaluate this elasticity, we anchor the optimal technique to a baseline rate of capital accumulation. Following the normal reproduction path, the normal accumulation rate is driven by the normal profit rate and the normal recapitalization share, net of depreciation:

$$g_K^N = \chi^N r^N - \delta \tag{14}$$

Expressed dynamically relative to this normal path, the aggregate transformation elasticity becomes:

$$\theta(\omega) = 1 + \frac{b^N(\omega)}{g_K^N} = 1 + \frac{a^N(\omega) - q^*(\omega)}{g_K^N} \tag{15}$$

3.9 Proposition 2. *Endogeneity of the transformation elasticity.* Under Assumption 1, the transformation elasticity θ is not an exogenous technical constant. It emerges endogenously from the cost-minimizing choice of technique. When distributive pressure forces firms to alter mechanization via (9), it simultaneously shifts the structural efficiency of accumulation via (15). The elasticity θ is therefore a function of the wage share ω, the frontier parameters $\lambda_0, \lambda_1, \lambda_2$, and the normal accumulation rate g_K^N.

3.2 Heterogeneous Capital, the Mechanization Gap, and State-Dependent Viability

3.10 Mainstream macroeconomics treats the capital stock as a homogeneous aggregate, abstracting from the internal architecture of the firm. Following the theory of the divided firm (Vidal, 2015, 2019), we unbundle aggregate productive capital into extensive nonresidential structures (K^{NRC}) and intensive machinery (K^{ME}). These components do not enter the labor process symmetrically.

Structures form the passive infrastructure envelope—the factory floor and logistics shell. Machinery is the active component that restructures the production of relative surplus value (Marx, 1999, Vol. I, Ch. 15).

3.11 Assumption 2. *Heterogeneous capital decomposition.* Aggregate productive capital decomposes as $K_t^{cap} = K_t^{NRC} + K_t^{ME}$. Nonresidential structures evolve independently of short-run distributive conflict (ω) and follow a normal accumulation corridor governed by long-run reproduction conditions:

$$g^N_{K^{NRC},t} = \chi^N r_t^N - \delta \tag{16}$$

This extensive envelope $\Gamma(K^{NRC})$ bounds the space of tactical technique choice.

3.2.1 The Extensive Envelope and the Intensive Margin

3.12 Given the fixed infrastructural envelope, the **intensive margin** of accumulation is the tactical choice of the mechanization gap, defined in natural logs as:

$$\kappa_t = k_t^{ME} - k_t^{NRC} \tag{17}$$

This gap measures mechanization intensity relative to the extensive plant scale. The capacity identity decomposes potential output into the extensive envelope and the intensive capacity yield Ψ_t:

$$y_t^p = \theta_{NRC} k_t^{NRC} + \Psi_t(\kappa_t, \omega_{t-1}, p_t^\kappa) \tag{18}$$

The intensive yield Ψ_t is not a fixed engineering constant. It is the outcome of a decentralized cost-minimization problem governed by the regime of accumulation, as formalized below.

3.2.2 Cost-Surplus Maximization and the Okishio Viability Yield

3.13 The firm chooses the mechanization gap κ_t to maximize the net viability surplus, subject to the regime-conditioned mechanization frontier:

$$\Phi(\kappa_t) = \lambda_0 + \lambda_1 \kappa_t + \lambda_2 \kappa_t^2 \tag{19}$$

The *realized* capacity yield is governed by the **Marx–Okishio viability condition**: a cost-minimizing firm deploys a new technique only if it reduces unit costs at current distributive and price parameters (Okishio, 1961; Basu, 2022). We formalize this as a cost-surplus maximization problem:

$$\max_{\kappa_t} \; c(\kappa_t) = \Phi(\kappa_t) - \kappa_t \cdot MC(\omega_{t-1}, p_t^\kappa) \tag{20}$$

3.14 The marginal viability cost $MC(\omega_{t-1}, p_t^\kappa)$ is determined by two channels:

1. **The distributive dividend (labor saving).** Increasing κ_t substitutes machinery for labor. The value of this displacement is proportional to the lagged wage share (ω_{t-1}), reflecting the wage settlement period prior to current capacity realization. A higher wage share *reduces* the net cost of mechanization.

2. **The relative price penalty (capital cost).** Increasing κ_t requires accumulating machinery faster than structures. The cost of this composition shift is proportional to the log relative

price of machinery ($p_t^\kappa = \ln P_t^{ME} - \ln P_t^{NRC}$). A higher relative price *increases* the marginal cost.

We specify the marginal viability cost as:

$$MC(\omega_{t-1}, p_t^\kappa) = \gamma_0 - \gamma_1 \omega_{t-1} + \gamma_2 p_t^\kappa \tag{21}$$

where $\gamma_1 > 0$ captures the wage-savings dividend and $\gamma_2 > 0$ captures the relative price penalty.

3.15 Substituting (19) and (21) into (20) and solving the first-order condition yields the optimal mechanization gap:

$$c(\kappa_t) = \lambda_0 + \left[\lambda_1 - (\gamma_0 - \gamma_1 \omega_{t-1} + \gamma_2 p_t^\kappa)\right] \kappa_t + \lambda_2 \kappa_t^2 \tag{22}$$

$$\frac{dc}{d\kappa_t} = \lambda_1 - \gamma_0 + \gamma_1 \omega_{t-1} - \gamma_2 p_t^\kappa + 2\lambda_2 \kappa_t^* = 0 \tag{23}$$

$$\kappa_t^*(\omega_{t-1}, p_t^\kappa) = \frac{(\gamma_0 - \lambda_1) - \gamma_1 \omega_{t-1} + \gamma_2 p_t^\kappa}{2\lambda_2} \tag{24}$$

3.16 Proposition 3. *Dual-channel induced innovation.* Under Assumption 1 (concavity) and Assumption 2 (heterogeneous capital), the optimal mechanization gap satisfies:

$$\frac{\partial \kappa_t^*}{\partial \omega_{t-1}} = -\frac{\gamma_1}{2\lambda_2} > 0 \tag{25}$$

$$\frac{\partial \kappa_t^*}{\partial p_t^\kappa} = \frac{\gamma_2}{2\lambda_2} < 0 \tag{26}$$

A rising wage share forces cost-minimizing firms to widen the mechanization gap, substituting machinery for labor. A rising relative price of machinery suppresses the optimal gap, as the structural cost of the composition shift penalizes mechanization. *Proof.* Immediate from differentiating (24). ■

3.2.3 The Marginal Capacity Payoff and the Econometric Bridge

3.17 The macroeconomic translation from the mechanization gap to potential capacity is given by the **marginal capacity payoff** (MP_κ), the derivative of the intensive capacity yield Ψ_t with respect to κ_t. By the envelope theorem:

$$MP_\kappa = \frac{\partial \Psi_t}{\partial \kappa_t} = \frac{\partial \Phi}{\partial \kappa_t} - MC(\omega_{t-1}, p_t^\kappa) \tag{27}$$

$$= (\lambda_1 + 2\lambda_2 \kappa_t) - (\gamma_0 - \gamma_1 \omega_{t-1} + \gamma_2 p_t^\kappa) \tag{28}$$

$$= \underbrace{(\lambda_1 - \gamma_0 + 2\lambda_2 \kappa_t)}_{\text{autonomous yield}} + \gamma_1 \omega_{t-1} - \gamma_2 p_t^\kappa \tag{29}$$

3.18 Integrating the marginal payoff with respect to κ_t (treating ω_{t-1} and p_t^κ as predetermined shifters) and defining θ_κ, $\theta_{\omega\kappa} \equiv \gamma_1$, and $\theta_{\kappa p\kappa} \equiv -\gamma_2$ yields the intensive capacity yield function:

$$\Psi_t = \theta_\kappa \kappa_t + \theta_{\omega\kappa}(\omega_{t-1}\kappa_t) + \theta_{\kappa p\kappa}(p_t^\kappa \kappa_t) \tag{30}$$

Substituting (30) back into (18) recovers the structural capacity equation:

$$y_t^p = \theta_{NRC} k_t^{NRC} + \theta_\kappa \kappa_t + \theta_{\omega\kappa}(\omega_{t-1}\kappa_t) + \theta_{\kappa p\kappa}(p_t^\kappa \kappa_t) \tag{31}$$

The interaction terms are not ad-hoc econometric additions. They are the macroeconomic footprint of the divided firm's Okishio viability constraint. Distributive pressure and relative price enter because they alter the marginal capacity payoff of the mechanization gap.

3.2.4 The Aggregate Transformation Elasticity

3.19 The aggregate transformation elasticity (θ_t^{SpecB}) is the ratio of the total differentials of potential capacity and aggregate log capital along the historical accumulation path. The total differential of (31), holding the slow-moving state variables fixed, separates the extensive envelope from the intensive margin:

$$dy_t^p = \theta_{NRC} dk_t^{NRC} + MP_\kappa d\kappa_t \tag{32}$$

$$= \theta_{NRC} dk_t^{NRC} + (\theta_\kappa + \theta_{\omega\kappa}\omega_{t-1} + \theta_{\kappa p\kappa} p_t^\kappa)\, d\kappa_t \tag{33}$$

3.20 The variation in aggregate capital follows from the Divisia index of the heterogeneous components. Since $s_{ME,t} + s_{NRC,t} = 1$ and $d\kappa_t = dk_t^{ME} - dk_t^{NRC}$:

$$dk_t^{cap} \approx s_{ME,t} dk_t^{ME} + s_{NRC,t} dk_t^{NRC} \tag{34}$$

$$= dk_t^{NRC} + s_{ME,t} d\kappa_t \tag{35}$$

Dividing (33) by (35) yields the elasticity along the empirical growth path:

$$\theta_t^{SpecB} = \frac{\theta_{NRC} + (\theta_\kappa + \theta_{\omega\kappa}\omega_{t-1} + \theta_{\kappa p\kappa} p_t^\kappa)\, \frac{d\kappa_t}{dk_t^{NRC}}}{1 + s_{ME,t} \frac{d\kappa_t}{dk_t^{NRC}}} \tag{36}$$

3.21 The aggregate transformation elasticity θ_t^{SpecB} in (36) is state-dependent: its value shifts with the distributive channel (ω_{t-1}), the price channel (p_t^κ), and the empirical growth path $d\kappa_t / dk_t^{NRC}$. Whether θ_t^{SpecB} exceeds or falls below unity is an empirical question determined by the interaction of these channels under a given accumulation regime. The identification strategy in Section 3.4 specifies how this regime-dependence is estimated and tested.

3.3 The Peripheral Extension: Balance-of-Payments Constraint and the Corner Solution

3.22 In the capitalist periphery, the intensive margin of accumulation is structurally tethered to the world market and constrained by domestic social friction. Following Prebisch (1981) and Kaldor (1959), peripheral capitalism is characterized by high consumption propensities out of profits, structural inelasticities in agriculture, and a chronic vulnerability to balance-of-payments crises (Ahumada, 2019). This introduces a dual realization bottleneck: the macroeconomic capacity to import and the inelastic domestic supply of wage-goods. Section 2.5 established this constraint qualitatively as the peripheral accumulation wedge. This subsection formalizes it as a strict

macroeconomic boundary constraint on the cost-minimizing firm's choice of technique, integrating the structuralist insights of Prebisch (1981) into the heterogeneous capital framework of Section 3.2.

3.3.1 The Import-Biased Intensive Margin and the Balance of Payments Constraint

3.23 In Section 3.2, the cost-minimizing firm chooses the mechanization gap κ_t to maximize the net viability surplus:

$$\max_{\kappa_t} \; c(\kappa_t) = \Phi(\kappa_t) - \kappa_t \cdot MC(\omega_{t-1}, p_t^{\kappa}) \tag{37}$$

where $\Phi(\kappa_t) = \lambda_0 + \lambda_1\kappa_t + \lambda_2\kappa_t^2$ is the regime-conditioned mechanization frontier, and $MC(\omega_{t-1}, p_t^{\kappa}) = \gamma_0 - \gamma_1\omega_{t-1} + \gamma_2 p_t^{\kappa}$ is the marginal viability cost of the gap. In the core, this optimization proceeds unconstrained, yielding the optimal mechanization gap κ_t^*. In the periphery, however, active mechanization is strictly import-biased: widening the mechanization gap requires imported capital goods, specialized machinery, and technologically non-substitutable spare parts (Vernengo, 2006).

3.24 Assumption 3. *Import-biased mechanization.* The physical import requirement per unit of output (m_t^{req}) is a linear function of the chosen technique:

$$m_t^{req} = \nu \cdot \kappa_t \tag{38}$$

where $\nu > 0$ captures the structural import-intensity of the mechanization gap. Simultaneously, the physical capacity to satisfy this requirement is bounded by the macroeconomic **capacity to import** per unit of output ($m_t^{avail} = M_t^{avail}/Y_t$). Following Prebisch (1981), peripheral export growth is governed by the center's income elasticity of demand for primary goods ($\epsilon_{m,c} < 1$), structurally lagging behind domestic import demand. This external asymmetry, compounded by high consumption propensities out of capitalist profits (Kaldor, 1959), severely squeezes the foreign exchange available for productive accumulation. This imposes a strict external ceiling on the choice of technique:

$$\nu \cdot \kappa_t \leq m_t^{avail} \implies \kappa_t \leq \bar{\kappa}_t \equiv \frac{m_t^{avail}}{\nu} \tag{39}$$

3.3.2 The Macroeconomic Allocation of Foreign Exchange

3.25 Assumption 4. *Balance-of-payments constraint.* Total import demand cannot exceed export capacity. The total supply of foreign exchange per unit of output is driven by primary exports ($x_t = p_t^x X_t/Y_t$). Due to the center's low income elasticity for primary goods, x_t is structurally rigid. The total demand for foreign exchange per unit of output consists of three competing components:

1. **Wage-goods imports** (m_t^W): driven by the wage share ω_{t-1} and the structural inelasticity of domestic agriculture (Kaldor, 1959). $m_t^W = m_w\omega_{t-1}$.
2. **Conspicuous consumption imports** (m_t^C): driven by the profit share π_t and the high propensity of the peripheral upper strata to imitate the consumption patterns of the center (Prebisch, 1981; Kaldor, 1959). $m_t^C = m_c c_k \pi_t$.
3. **Productive capital imports** (m_t^K): driven by the mechanization gap. $m_t^K = \nu\kappa_t$.

3.26 Assumption 5. *Surplus leakage.* The propensity to consume out of profits satisfies $c_k > c_k^{\min}$, where $c_k^{\min}$ is the threshold below which domestic savings suffice to finance capital formation without external borrowing. Under the developmentalist regime, the leakage channel is conspicuous consumption imports by domestic elites (Prebisch, 1981; Kaldor, 1959). Under the neoliberal regime, the leakage channel is profit repatriation by multinational corporations and capital flight (Ahumada, 2019). In both cases, $c_k > c_k^{\min}$ ensures that the effective profit share available for productive reinvestment is compressed below the nominal level: $\pi_t^{eff} = \pi_t \cdot \phi_t < \pi_t$.

3.27 The balance of payments constraint requires:

$$m_w \omega_{t-1} + m_c c_k \pi_t + \nu \kappa_t \leq x_t \tag{40}$$

Using the distributive identity $\pi_t = 1 - \omega_{t-1}$, we substitute and solve for the maximum feasible mechanization gap:

$$m_w \omega_{t-1} + m_c c_k (1 - \omega_{t-1}) + \nu \kappa_t \leq x_t \tag{41}$$

$$\nu \kappa_t \leq x_t - m_c c_k + (m_c c_k - m_w)\omega_{t-1} \tag{42}$$

$$\bar{\kappa}_t(\omega_{t-1}) = \frac{1}{\nu}\left[x_t - m_c c_k + (m_c c_k - m_w)\omega_{t-1}\right] \tag{43}$$

This equation formalizes the structuralist critique: the base capacity to import capital goods is determined by export earnings minus the baseline luxury consumption of the capitalist class $(x_t - m_c c_k)$. The sign of $(m_c c_k - m_w)$ determines whether a rising wage share expands or contracts the feasible mechanization gap.

3.3.3 The Constrained Cost-Surplus Maximization Problem

3.28 The peripheral firm's tactical problem is therefore a constrained cost-surplus maximization:

$$\max_{\kappa_t} \; c(\kappa_t) = \Phi(\kappa_t) - \kappa_t \cdot MC(\omega_{t-1}, p_t^{\kappa}) \tag{44}$$

$$\text{subject to:} \quad \kappa_t \leq \bar{\kappa}_t \tag{45}$$

This formulation defines two distinct accumulation regimes, endogenous to the interaction between domestic distributive conflict and the global market.

3.3.4 Endogenous Regime Shifts: The Corner Solution

3.29 The firm's unconstrained optimal technique $\kappa_t^*(\omega_{t-1})$ is strictly increasing in the wage share $(\partial \kappa_t^* / \partial \omega_{t-1} > 0)$ due to the Okishio labor-saving bias established in (25).

3.30 Proposition 4. *Peripheral regime shifts.* Two regimes are endogenous to the interaction between domestic distributive conflict and the global market:

1. **Regime A (Slack BoP):** When $\kappa_t^* \leq \bar{\kappa}_t$, the firm solves the standard first-order condition, yielding κ_t^*. The labor-saving bias successfully extracts the Okishio distributive dividend.

2. **Regime B (Peripheral realization trap):** When $\kappa_t^*(\omega_{t-1}) > \bar{\kappa}_t(\omega_{t-1})$, the constraint binds and the firm is forced into a corner solution:

$$\kappa_t = \bar{\kappa}_t \tag{46}$$

The corner solution arises through two intersecting structural vulnerabilities. First, if $m_c c_k$ is large (the conspicuous consumption drain), the baseline FX availability is severely squeezed, lowering the intercept of $\bar{\kappa}_t$. Second, if domestic agriculture is highly inelastic ($m_w > m_c c_k$), a rising wage share simultaneously increases the *demand* for capital imports (via κ_t^*) while *contracting* the supply of available FX (via $\bar{\kappa}_t$). *Proof.* Immediate from the Kuhn–Tucker conditions applied to (44)–(45). ■

3.3.5 Effective Profit Compression and the Blocked Surplus

3.31 Definition 3. *FX-Conversion Ratio.* The FX-conversion ratio ϕ_t is the ratio of the maximum feasible technique to the optimal unconstrained technique:

$$\phi_t = \min\left(1, \frac{\bar{\kappa}_t}{\kappa_t^*}\right) \tag{47}$$

When the external constraint binds ($\kappa_t^* > \bar{\kappa}_t$), $\phi_t < 1$. This generates a direct compression of the effective profit share available for intensive accumulation.

3.32 Definition 4. *Blocked surplus.* Even if domestic wage suppression yields a high nominal profit share (π_t), the physical inability to convert that surplus into imported machinery squeezes the effective profit rate:

$$\pi_t^{eff} = \pi_t \cdot \phi_t = (1 - \omega_{t-1}) \cdot \phi_t \tag{48}$$

The difference constitutes the **blocked surplus**:

$$\text{Blocked Surplus} = \pi_t(1 - \phi_t) \tag{49}$$

Because the accumulation of machinery is driven by the *effective* profit share ($g_{K^{ME}} = \chi^{ME}\pi_t^{eff} - \delta^{ME}$), the constraint decouples nominal profitability from capacity formation. If capitalists squeeze wages to raise π_t, they inadvertently increase conspicuous consumption (m_t^C), which lowers $\bar{\kappa}_t$ and compresses ϕ_t.

3.3.6 The Regime-Shifting Aggregate Transformation Elasticity

3.33 Definition 5. *BoP binding ratio.* The BoP binding ratio τ_t is the ratio of productive capital imports to import capacity:

$$\tau_t \equiv \frac{\nu \cdot I_t^{ME}}{\text{ToT}_t \times X_t} \tag{50}$$

where $I_t^{ME} = K_t^{ME} - (1 - \delta^{ME})K_{t-1}^{ME}$ is gross investment in machinery and equipment, derived from the perpetual inventory identity, and $\text{ToT}_t \times X_t$ proxies the import capacity (FX supply). Under Assumption 3, productive capital imports $m_t^K = \nu\kappa_t$ are proxied empirically by gross investment in machinery and equipment I_t^{ME}, since mechanization is import-biased in the periphery. The regime switch is governed by τ_t:

- $\tau_t < \bar{\tau}$: BoP constraint is **slack** (Regime A: unconstrained optimum).
- $\tau_t \geq \bar{\tau}$: BoP constraint is **binding** (Regime B: corner solution).

The threshold $\bar{\tau}$ is determined empirically by the threshold regression in Section 4.

3.34 The peripheral extension generates a regime-shifting aggregate transformation elasticity. Under **Regime A** (slack BoP), the constraint does not bind, and the elasticity coincides with the unconstrained expression from Section 3.2:

$$\theta_t^A = \theta_t^{SpecB} = \frac{\theta_{NRC} + (\theta_\kappa + \theta_{\omega\kappa}\omega_{t-1} + \theta_{\kappa p\kappa} p_t^\kappa)\,\frac{d\kappa_t^*}{dk_t^{NRC}}}{1 + s_{ME,t}\frac{d\kappa_t^*}{dk_t^{NRC}}} \tag{51}$$

3.35 Under **Regime B** (binding BoP), the firm is forced into the corner solution $\kappa_t = \bar{\kappa}_t$. The marginal capacity payoff at the corner is strictly positive because the firm would like to mechanize further but cannot:

$$MP_\kappa\big|_{\kappa=\bar{\kappa}} = \Phi'(\bar{\kappa}_t) - MC(\omega_{t-1}, p_t^\kappa) > 0 \tag{52}$$

The Kuhn–Tucker multiplier on (45) is strictly positive, confirming that the constraint binds. The aggregate transformation elasticity under the binding regime becomes:

$$\theta_t^B = \frac{\theta_{NRC} + MP_\kappa\big|_{\kappa=\bar{\kappa}} \cdot \frac{d\bar{\kappa}_t}{dk_t^{NRC}}}{1 + s_{ME,t}\frac{d\bar{\kappa}_t}{dk_t^{NRC}}} \tag{53}$$

Since $\bar{\kappa}_t$ is determined by the BoP constraint (43) rather than by the first-order condition (23), the growth path $d\bar{\kappa}_t/dk_t^{NRC}$ differs from the unconstrained path $d\kappa_t^*/dk_t^{NRC}$. The FX-conversion ratio $\phi_t = \bar{\kappa}_t/\kappa_t^* < 1$ compresses the intensive margin, and the elasticity under the binding regime is strictly lower:

$$\theta_t^B < \theta_t^A \tag{54}$$

3.36 Proposition 5. *Regime-shifting elasticity.* The peripheral aggregate transformation elasticity is a regime-dependent parameter:

$$\theta_t^{periphery} = \theta_t^A \cdot \mathbf{1}\{\kappa_t^* \leq \bar{\kappa}_t\} + \theta_t^B \cdot \mathbf{1}\{\kappa_t^* > \bar{\kappa}_t\} \tag{55}$$

The elasticity shifts discretely when the BoP constraint becomes binding. The regime switch is governed by the interaction between the desired mechanization gap $\kappa_t^*(\omega_{t-1}, p_t^\kappa)$ and the maximum feasible gap $\bar{\kappa}_t(\omega_{t-1})$ determined by export capacity, conspicuous consumption, and wage-goods imports. An increase in the wage share ω_{t-1} simultaneously raises κ_t^* (via the Okishio labor-saving bias) and can contract $\bar{\kappa}_t$ (via the wage-goods import channel when $m_w > m_c c_k$), making the regime switch more likely. This is the peripheral realization trap: distributive pressure that would induce productive mechanization in the center instead triggers the BoP constraint in the periphery, truncating the transformation elasticity.

3.3.7 The Failure of the Okishio Mechanism and Structural Inflation

3.37 Proposition 6. *Blocked Okishio mechanism.* At the boundary $\kappa_t = \bar{\kappa}_t$, the implementation of the Okishio viability condition is blocked. The firm satisfies the viability criterion—the technique would reduce unit costs—but cannot deploy it because the required machinery cannot be imported. The wage-savings dividend cannot be capitalized into physical capacity. The firm suffers a persistent shortfall in its cost-surplus relative to its unconstrained optimal state. *Proof.* At $\kappa_t = \bar{\kappa}_t$, the Kuhn–Tucker multiplier on (45) is strictly positive, so the first-order condition $\partial c/\partial \kappa_t = 0$ does not hold. The marginal viability yield exceeds the marginal cost at the boundary, but the firm cannot expand further. ■

3.38 Unable to extract the distributive dividend via mechanization, capitalists attempt to recover the blocked surplus through the only remaining channel: price markups. When the wage-goods bottleneck forces nominal wage adjustments and the external BoP bottleneck blocks defensive mechanization, any further increase in the wage share directly crushes profitability. The resulting defensive pricing behavior ignites the structural inflation dynamic (Sunkel, 1958; Prebisch, 1981). This is the peripheral realization trap: nominal profitability coexists with truncated mechanization and stalled structural envelopes, permanently destabilizing the long-run coupling between plant and machinery.

3.4 Synthesis: What the Three Models Jointly Demonstrate

3.38 The three models developed in this section formalize a single theoretical claim: that decentralized, individually cost-minimizing decisions generate aggregate disproportionality under the unbalanced growth closure (Okishio, 1961; Basu, 2026). Table 1 summarizes the articulation between the Okishio and Vidal frameworks that underpins this claim.

Table 1: Theoretical Coherence: Market Anarchy and the Internally Divided Firm Articulation Across Three Models

Element	**Descentralized Accumulation**	**Internally Divided Firm**	**Formal model captures**
Micro logic	Viability condition (cost reduction at prevailing prices)	Internal contradiction (control vs. cooperation)	Optimization problem = cost-minimizing pole; constraints = cooperation limits
Macro logic	Decentralized decisions → disproportionality	Divided firm → contested operating conditions	$\theta \neq 1$ captures gap between accumulation and capacity
Disequilibrium	Unbalanced growth (supply ≠ demand)	"Mediocre sufficiency" (Vidal, 2022) (no coherent optimum)	μ demonstrates gap; θ captures structural inefficiency
Peripheral specificity	BoP constraint blocks *implementation* of viability	External constraint compounds internal division	Corner solution: $\kappa = \bar{\kappa}$ when constraint binds

3.39 The formal model does not assert equilibrium. It demonstrates the negative: that capitalist accumulation, left to its own decentralized logic, does not generate proportional reproduction. The capacity utilization rate μ is the empirical signature of this disproportionality: it shows that the

normal operating conditions of capitalist firms, determined by their cost-minimizing mechanization choices, do not meet aggregate demand conditions. The transformation elasticity θ captures this structural gap between accumulation and realized capacity. Whether θ exceeds or falls below unity is an empirical question determined by the interaction of distributive and price channels under a given accumulation regime. In the center, this gap is cyclical—mediated by institutional configurations that can be exhausted and reconstituted (Vidal, 2015, 2019). In the periphery, this gap is permanent—locked in by the structural ceiling of external subordination (Prebisch, 1981; Ahumada, 2019).

3.40 Section 4 consolidates the empirical identification strategy. It translates the three analytical models into estimable cointegrating specifications, derives the threshold regression for the peripheral corner solution, and specifies the capacity recovery protocol. The empirical analysis bounds this multi-sector mechanism to a single-sector aggregate, which masks internal sectoral divergences but preserves the core transformation logic.

4 Data, Measurement, and Empirical Results

4.1 Data and Measurement

4.1.1 The Dual Valuation Distinction

This section constructs the capital stock series for the choice-of-technique models of **??**. The BEA Fixed Assets Accounts report a single current-cost net stock. That series treats the physical capacity of installed capital and the monetary value of capital advanced as one magnitude, and the conflation introduces a spurious correlation between capital intensity and the wage share: as mechanization raises capital intensity, Consumption of Fixed Capital (CFC) rises in the GVA denominator, depressing GVA-basis wage shares even when the rate of surplus-value extraction is unchanged. To eliminate this bias, I reconstruct the stock account-by-account using the Generalized Perpetual Inventory Method (GPIM) of Shaikh (2016, App. 6.5), which builds capacity and profitability measures from direct nominal investment flows and asset-specific price indexes.

The dual-valuation distinction. Estimates of capacity utilization under my unbalanced growth framework, require distinct capital stock variables. The capital value initially invested in plant and equipment returns gradually to its money form as fixed assets depreciate; the accumulated depreciation allowances remain part of total capital advanced regardless of their monetary form, because their recovery is what permits the reproduction of the capitalist enterprise (Shaikh, 2016, App. 6.3). I therefore distinct two different types of measurements of capital stocks:

1. **Capital stock for capacity estimation: gross real GPIM stock.** Productive capacity depends on capital goods still physically in operation. The capital stock measure adequate for estimating capacity is the gross real stock $K^{G,R}_{j,t}$, weighted exclusively by a physical survival schedule. Productive capacity Y^p_t is itself a latent variable, recovered through the estimation of the transformation elasticity θ (**??**); the gross real GPIM stock is the accumulation variable k_t included in my estimation equation, not a measure of capacity itself. A machine whose book value has fallen to zero, but remains in service until it is physically retired, still should be accounted as physical capital in operation. The OECD manual confirms that the gross stock,

valued at "as new" prices, "is widely used as a broad indicator of the productive capacity of a country" (OECD, 2001).

2. **Profitability measure: net current-cost stock.** Profitability is a monetary ratio between a profit flow and the value of capital advanced. The relevant measure is the net current-cost stock $K^{N,V}_{j,t}$, which tracks the monetary value of capital after financial depreciation and write-downs. This measure serves as the anchor for the estimation of an implicit price of capital goods under a stock-flow-consistent framework but is not included within the structural estimation of non-observed productive capacities.

A single aggregate net stock serving both purposes is common in empirical macroeconomics. This practice introduces a mechanical correlation between the capital stock and the depreciation flow, introducing potential biases to identify distributive effects on productive capacities formation.

Valuation choice within the GPIM. The dual-valuation distinction does not require imposing a single valuation convention throughout. Shaikh (2016, App. 6.3) advocates the gross stock as the denominator for both the profit rate and the capacity ratio. The GPIM he develops, however, constructs the survival schedule $S_j(v)$, the age-price schedule $(1-d_j)^v$, and the SFC implicit price index $P_{j,t}$ as distinct components within a single stock-flow-consistent recursion. This structure permits the researcher to select the valuation for each analytical purpose without abandoning accounting coherence.

For the profit rate, the denominator must measure the monetary value of capital advanced. The net current-cost stock serves this role: it tracks the book value of capital that the firm must recover to continue production. A machine that has depreciated to a low book value may still operate at full physical capacity, but the firm's cost-recovery problem is defined by the value it must recoup, not by the physical state of the equipment (Basu, 2013). For estimates related to capacity utilization, the capital stock input must reflect the physical capital still in operation. The gross real GPIM stock, weighted exclusively by the survival schedule $S_j(v)$ and excluding the age-price factor $(1-d_j)^v$, serves this role.

The quantitative significance of the gross/net distinction is not confined to the center. Ffrench-Davis and Vivanco (2016) show that in Chile depreciation absorbs between 8 and 13 percent of GDP and that net investment is barely half of gross investment in the most recent period. Service lives of machinery have declined from 20 years in the 1940s to 11 years by 2005, reflecting accelerating obsolescence. They conclude that the capital stock entering a production function must be the gross stock of physically surviving assets, not the depreciated monetary residual.

4.1.2 Gross Capital Stock Reconstruction and Measurement in the United States

The Generalized Perpetual Inventory Method. In this part I present the procedures of construction for the capital stock series for the United States. Machinery (ME) and nonresidential structures (NRC) do not share a natural physical unit; they become comparable only through a monetary-mediated accounting procedure. I reconstruct the stock account-by-account rather than aggregating nominal values and deflating with a generic index. The GPIM accumulation rules for historical-cost, current-cost, and constant-dollar stocks (Shaikh, 2016, App. 6.5) are derived in detail in Appendix **??**; here I state the construction.

Physical survival of capital vintages follows a Weibull retirement distribution. For asset class $j \in \{\text{ME}, \text{NRC}\}$, the proportion of a vintage surviving after v years is:

$$S_j(v) = \exp\left[-\left(\frac{v}{\lambda_j}\right)^{\alpha_j}\right] \tag{56}$$

where α_j is the Weibull shape parameter and the scale parameter is $\lambda_j = L_j/\Gamma(1 + 1/\alpha_j)$, derived from the mean service life L_j. Unlike geometric decay models that continuously erode the physical stock, the Weibull curve ensures that capital goods remain fully operative until they are physically removed from service. Empirical studies of directly observed asset discards confirm that the Weibull family provides a flexible approximation of retirement patterns. Nomura (2005), using Japanese micro-data on 66 asset classes, finds that almost half of assets exhibit a progressively increasing hazard rate ($\alpha > 1$). The calibrated parameters $L_{\text{ME}} = 14$, $\alpha_{\text{ME}} = 1.7$ and $L_{\text{NRC}} = 30$, $\alpha_{\text{NRC}} = 1.6$ follow the BEA Fixed Assets methodology (Fraumeni et al., 1997).

Physical survival alone cannot identify the SFC implicit price index. The BEA current-cost net stock anchors reflect both physical retirement and age-price decay. To recover the asset-specific price index, I impose an externally documented geometric declining-balance age-price profile:[2]

$$V_j(v) = S_j(v) \cdot (1 - d_j)^v \tag{57}$$

where d_j is the asset-specific age-price decay rate. The declining-balance formula applied to the calibrated service lives yields $d_{\text{ME}} = 0.110$ and $d_{\text{NRC}} = 0.024$, consistent with the geometric rates documented by the BEA (Fraumeni et al., 1997).

The SFC implicit price index $P_{j,t}$ is the price level that closes the stock-flow identity for the net current-cost stock. In each period, the BEA anchor must satisfy

$$K_{j,t}^{\text{NET,CC}} = I_{j,t}^{\text{NOM}} + \frac{P_{j,t}}{100} \sum_{v=1}^{L_j} V_j(v)\, I_{j,t-v}^{\text{REAL}} \tag{58}$$

where $I_{j,t-v}^{\text{REAL}} = I_{j,t-v}^{\text{NOM}}/(P_{j,t-v}/100)$ is real investment deflated by the previously recovered SFC price. The first term on the right-hand side is current-year nominal investment; the second is the net-value-weighted sum of surviving real vintages revalued at the current price. The price index is "implicit" because it is not directly observed: it is the unique value that, applied to the survival-weighted real vintages, reproduces the BEA's reported current-cost net stock. Solving (58) for $P_{j,t}$:

$$P_{j,t} = 100 \cdot \frac{K_{j,t}^{\text{NET,CC}} - I_{j,t}^{\text{NOM}}}{\displaystyle\sum_{v=1}^{L_j} V_j(v)\, I_{j,t-v}^{\text{REAL}}} \tag{59}$$

The forward recursion proceeds period by period: given $P_{j,t-1}, P_{j,t-2}, \ldots$, the current price $P_{j,t}$ is uniquely determined by the requirement that the constructed stock reproduces the BEA anchor. Appendix **??** documents the recursion, the initial conditions, and the base-year normalization. The

[2]The BEA's current methodology employs geometric depreciation, which assumes that the efficiency pattern and the depreciation pattern have the same form (Fraumeni et al., 1997). The GPIM construction separates these two dimensions: physical survival follows a Weibull retirement distribution, while financial age-price decay follows a geometric profile. This separation is necessary to construct distinct capacity and profitability measures.

BEA's Fisher quantity index is retained solely as a validation comparison and does not enter the GPIM baseline.

The capital stock used to estimate productive capacity is the gross real GPIM stock, constructed as the survival-weighted sum of real investment vintages:

$$K_{j,t}^{G,R} = \sum_{v=0}^{L_j} S_j(v)\, I_{j,t-v}^{\text{REAL}} \tag{60}$$

where $I_{j,t-v}^{\text{REAL}}$ is real investment deflated by the previously recovered SFC price. Because eq. (60) relies exclusively on the survival schedule $S_j(v)$ and excludes the age-price factor $(1-d_j)^v$, it measures the constant-price purchasing power embodied in heterogeneous capital goods still in physical operation. Financial write-downs, revaluation gains, and depreciation allowances do not enter.

Validation and sample scope. The GPIM construction closes the stock-flow identity exactly: the maximum absolute reconstruction residual across both asset classes is 10^{-6} million dollars, confirming that the recovered SFC implicit price indexes reproduce the BEA current-cost net stock anchors. All GPIM series are normalized to 2017 = 100. The capital stock series has a 30-year warm-up period (1901–1930). The GPIM stock at any year t requires investment vintages extending back L_j years. Because the nominal investment series begins in 1901, the vintage summation for the longest-lived asset class (nonresidential structures, $L_{\text{NRC}} = 30$) is truncated for any year before 1931. The stock values for 1901–1930 are therefore incomplete, omitting pre-1901 vintages that would still be physically in service. I restrict the estimation sample to 1931–2024, where the full 30-year vintage window is populated, yielding $T = 94$ annual observations. The sectoral scope is restricted to nonfinancial corporate (NFC) business, excluding non-profit institutions serving households (NPISH), government, and owner-occupied housing. This sectoral restriction isolates the private productive business sector from public infrastructure and imputed residential rents.

4.1.3 Distributive Conflicts and Wage Share Dynamics in the United States

4.1.4 Capital Series Harmonization and Historical Sources for Chile

This subsection describes the stock-flow consistent (SFC) methodology used to reconstruct gross and net capital stocks for Chile, 1900–1994, in constant 2003 Chilean pesos (CLP). The procedure enforces the accumulation identities for gross and net stocks jointly, yields implicit depreciation and retirement rates for each asset class, and delivers coherent series for machinery and equipment (ME), nonresidential construction (NRC), residential construction (RC), and their aggregates. The construction proceeds from a single investment control total, disaggregated by asset, accumulated via Weibull survival functions, and linked to net stocks through a gap-ratio approach anchored to the Hofman (2000b) overlap window.

The stock-flow identities. The reconstruction is governed by two laws of motion. The gross stock evolves according to

$$\Delta K_{j,t}^{g} = I_{j,t} - R_{j,t} \tag{61}$$

where $I_{j,t}$ is gross fixed investment and $R_{j,t}$ denotes physical retirements (scrapping of obsolete vintages, demolition, and technological withdrawal). The net stock evolves according to

$$\Delta K^n_{j,t} = I_{j,t} - D_{j,t} \tag{62}$$

where $D_{j,t}$ is depreciation (the monetary consumption of fixed capital). Combining (61) and (62) yields the joint SFC identity

$$\Delta K^g_{j,t} + R_{j,t} = \Delta K^n_{j,t} + D_{j,t} \tag{63}$$

which constrains the combination of depreciation and retirements at every t and for every asset j. Given observed stocks and investment, the implied flows are recovered as $D_{j,t} = I_{j,t} - \Delta K^n_{j,t}$ and $R_{j,t} = I_{j,t} - \Delta K^g_{j,t}$. The capital gap $B_{j,t} \equiv K^g_{j,t} - K^n_{j,t}$ measures the cumulative difference between installed capacity and book value.

The investment control total and the lambda bridge. The aggregate investment control total is drawn from the Pérez-Eyzaguirre (2019) reconstruction of gross fixed capital formation (GFCF), expressed in constant 2003 CLP. The control concept is machinery plus construction, explicitly excluding inventory variation.[3]

Pérez-Eyzaguirre report investment in a price space and asset aggregation that differ from the Hofman (2000a) benchmark. The two sources are linked by a level-normalization factor λ, which corrects for definitional differences between the two series rather than converting between price bases. The factor is estimated as the median of the annual ratio of Hofman to Pérez investment over the 95-year estimation window:

$$\lambda = \text{median}\left(\frac{I^{\text{Hofman}}_t}{I^{\text{Pérez}}_t}\right)^{1994}_{t=1900} = 4.1054 \tag{64}$$

The median rule is chosen for robustness to outliers in the early decades, where source quality is weaker. The scaled aggregate is then disaggregated into ME, NRC, and RC using normalized asset-specific investment shares from Hofman (2000a):

$$I_{j,t} = s_{j,t} \cdot \lambda \cdot I^{\text{Pérez}}_t, \qquad \sum_{j \in \{\text{ME, NRC, RC}\}} s_{j,t} = 1 \tag{65}$$

for all t.

Weibull survival and gross stock accumulation. Physical survival of capital vintages follows a Weibull retirement distribution. For asset class $j \in \{\text{ME}, \text{NRC}, \text{RC}\}$, the proportion of a vintage surviving after v years is

$$S_j(v) = \exp\left[-\left(\frac{v}{\lambda_j}\right)^{\alpha_j}\right] \tag{66}$$

where α_j is the Weibull shape parameter and the scale parameter is $\lambda_j = L_j/\Gamma(1 + 1/\alpha_j)$, derived from the mean service life L_j. The survival parameters are estimated from the Hofman (2000b) overlap window (1950–1994), in which both gross and net stocks are observed, by fitting Weibull

[3] Inventories are excluded from fixed-capital accumulation because the reconstruction models fixed capital only. Inventory investment does not enter the perpetual inventory recursion.

curves to the implied retirement patterns recovered from the stock-flow identities. Table 2 reports the calibrated parameters.

Table 2: Weibull survival parameters, Chile (1900–1994)

Asset	Scale (λ_j)	Shape (α_j)	Mean life (L_j, years)
ME	15.90	2.42	14.1
NRC	41.30	5.00	37.9
RC	51.16	5.00	47.0

All shape parameters exceed unity, indicating progressively increasing hazard rates: capital goods remain fully operative in early years and are withdrawn from service at an accelerating rate as they approach the mean service life. The shape parameters for NRC and RC are constrained at 5.0, reflecting the long and relatively uniform service lives of structures. The gross stock is accumulated as the survival-weighted sum of real investment vintages:

$$K^g_{j,t} = \sum_{v=0}^{L_j} S_j(v)\, I_{j,t-v} \tag{67}$$

Pre-1950 stocks are anchored to the observed Hofman levels at 1950 via a multiplicative factor that matches $K^g_{j,1950}$ by asset.

The gap-ratio approach and net stock construction. Unlike the U.S. GPIM, which recovers the net stock via a separate geometric age-price decay rate and an SFC implicit price recursion, the Chile reconstruction derives the net stock from the observed gross-net relationship in the Hofman (2000b) overlap window. The capital gap ratio

$$\gamma_j \equiv \frac{B_j}{K^g_j} = \frac{K^g_j - K^n_j}{K^g_j} \tag{68}$$

is computed by asset from the Hofman (2000b) data over the overlap period (1950–1994). Two regimes are distinguished: a "pre" ratio (median of years ≤ 1954) and a "post" ratio (median of years ≥ 1990), allowing for structural shifts in the gross-net relationship over the century. Outside the overlap window, the gap ratio is extended at the regime-specific median. The net stock is then recovered as

$$K^n_{j,t} = K^g_{j,t} \cdot (1 - \gamma_j) \tag{69}$$

This approach inherits the gross-net relationship observed in the Hofman accounting reference rather than imposing an external depreciation convention. It is appropriate for Chile because no independent age-price decay data are available for the historical period; the Hofman (2000b) overlap provides the only direct evidence on the joint behavior of gross and net stocks.

Implicit depreciation and retirement rates. Given the reconstructed stocks and investment flows, the implicit annual rates are defined as

$$z_{j,t} = \frac{R_{j,t}}{K^g_{j,t-1}}, \qquad \delta_{j,t} = \frac{D_{j,t}}{K^n_{j,t-1}} \tag{70}$$

where $z_{j,t}$ is the retirement (depletion) rate and $\delta_{j,t}$ is the depreciation rate. These rates are not imposed as parameters; they are derived from the data via the stock-flow identities (61)–(62). They serve as diagnostic indicators of capital durability and replacement dynamics, and are used in **??** to validate the internal consistency of the reconstruction.

Validation and estimation sample. The SFC construction is validated through four checks. First, additivity: all composite stocks and flows satisfy $K_{\text{NR}} = K_{\text{ME}} + K_{\text{NRC}}$ and $K_{\text{T}} = K_{\text{ME}} + K_{\text{NRC}} + K_{\text{RC}}$ at every t. Second, non-negativity: $K^g_{j,t} \geq 0$, $K^n_{j,t} \geq 0$, and $B_{j,t} \geq 0$ for all assets and years. Third, rate bounds: $z_{j,t} \in [0,1]$ and $\delta_{j,t} \in [0,1]$ for all observations from 1950 onward. Fourth, identity residuals: $\Delta K^g_{j,t} - (I_{j,t} - R_{j,t}) = 0$ and $\Delta K^n_{j,t} - (I_{j,t} - D_{j,t}) = 0$ to numerical precision. The estimation sample spans 1900–1994, yielding $T = 95$ annual observations. The overlap window (1950–1994) provides 45 years of direct Hofman (2000b) observations against which the reconstructed series are audited.

The BCCh forward extension (1995–2024). Extending the reconstruction beyond 1994 requires integrating the official national accounts published by the Central Bank of Chile (BCCh). A direct chain-linking of the BCCh series to the historical baseline would introduce a level discontinuity, because the two sources differ in base year, valuation conventions, and asset classification. To eliminate this discontinuity while preserving the validated 1900–1994 history, I employ a multiplicative splicing procedure that fixes the legacy panel in place.

For the overlap year 1994, an asset-specific bridge factor is computed as the ratio of the historical net stock to the BCCh reported net stock:

$$\beta_j = \frac{K^{n,\text{hist}}_{j,1994}}{K^{n,\text{BCCh}}_{j,1994}}, \quad j \in \{\text{ME, NRC, RC}\} \tag{71}$$

The same factor β_j is applied to both the BCCh net stock and the BCCh depreciation series for all years 1995–2024. This preserves the depreciation rates and asset dynamics implied by the BCCh while maintaining continuity with the constant 2003 CLP historical baseline. Table 3 reports the estimated bridge factors.

Table 3: Asset-specific bridge factors, 1994

Asset	Bridge factor (β_j)	Interpretation
ME	5.888	BCCh understates historical ME stock
NRC	1.033	Near parity
RC	1.050	Near parity

The large bridge factor for machinery and equipment reflects the substantially lower level at which the BCCh reports ME stocks relative to the Pérez-based historical reconstruction, a divergence driven by differences in asset coverage and valuation methodology. For nonresidential and residential construction, the bridge factors are close to unity, indicating that the two sources agree substantially on the level of structures.

Post-1994 real investment is inferred via the stock-flow identity:

$$I_{j,t} = \Delta K^n_{j,t} + D_{j,t}, \quad t \geq 1995 \tag{72}$$

where $D_{j,t}$ is the bridge-adjusted BCCh depreciation. The gross stock and physical retirements for the extended period are then reconstructed by applying the survival parameters from Table 2 to the complete 1900–2024 investment series via eq. (67). This ensures that the gross stock remains internally consistent across the splice boundary.

4.1.5 Distributive Conflicts and Wage Share Dynamics in Chile

The distributive regressor in the transformation elasticity is the wage share, ω_t. For Chile, the wage share is drawn from the revised dataset of real wages in Latin America constructed by **?** and updated by **?**. The dataset provides yearly series of real wages by skill level—unskilled (w_4), semi-skilled (w_3), and skilled (w_2)—together with an economically-active-population (EAP) weighted average (w), expressed in monthly purchasing-power-parity dollars at 1970 prices.[4]

The wage share is computed as the wage-to-income ratio $\omega_t = w_t/y_t$, where y_t is real income per worker derived from national income estimates. The functional distribution panel extends the **?** coverage to 2024 by splicing the revised wage series with post-2011 observations from official Chilean statistical sources. The resulting ω_t series spans 1920–2024, yielding $T = 105$ annual observations.

4.1.6 Aggregate Demand and Price Indices

The aggregate demand variables—real GDP (Y_t), exports FOB (X_t^{FOB}), and gross machinery investment ($I_{\text{ME},t}$)—are drawn from the aggregate demand reconstruction of Pérez-Eyzaguirre (2019), expressed in constant 2003 Chilean pesos. This series revises and extends the historical national accounts backbone of La República en Cifras (Lüders et al., 2016), correcting additivity and valuation problems in the original Clio-Lab series while preserving the sectoral GDP decomposition. Coverage spans 1860–2024.

The price and trade indices used to construct the relative capital price proxies (p_t^{κ}), the capacity to import (M_t^{cap}), are drawn from *La República en Cifras* (Lüders et al., 2016): the real exchange rate index, the US wholesale capital price index, and the terms of trade index. All three series are rebased to 2003 = 100 for internal consistency with the capital stock series.[5]

The capital import demand imbalance, combines sources from *La República en Cifras* with my data set of harmonized capital stocks through a log difference of $log(I_{MEt})$ - $log(M_t^{\text{cap}})$, which reconciles the supply export-side identified by the index of imports capacities (M_t^{cap}) of the external constraint, with the internal demand for foreign capital goods providing a both a more accurate variable for the regime shifting empirical strategy detailed below, while also a variant that allows me to check for roboustness of my estimates.

4.1.7 Balance of Payments and External Constraints in Chile

The external sector variables are drawn from the World Historical Balance of Payments Database (WBOP) constructed by Nievas and Piketty (2025), which provides annual series on global trade flows and the world balance of payments—including goods, services, income, and transfers—covering 57 core territories over the 1800–2025 period. Three series are used for Chile: the ratio of manufactured goods imports to GDP (a proxy for capital goods import dependence), net foreign income outflows as a share of GDP (a proxy for profit remittance drain), and the net current account balance as a

[4]The revised dataset is publicly available as supplementary material to **?**.

[5]The terms of trade index extends back to 1810; the real exchange rate and US wholesale price index cover 1900–2024.

share of GDP (an external balance indicator). All three series are expressed in current US dollars at market exchange rates and are available at the WBOP online repository.[6]

4.2 Econometric Strategy: Integrated Modified OLS and Cointegration

4.3 United States Empirical Evidence: Aggregate Capital Accumulation and Capacity Utilization

4.3.1 Baseline Empirical Estimates

4.3.2 Transformation Elasticity and Capacity Utilization Estimates

4.4 United States Empirical Evidence: Capital Heterogeneity and the Mechanization Gap

4.4.1 Heterogeneous Capital Stocks Empirical Estimates

4.4.2 Aggregate Transformation Elasticity and Capacity Utilization Estimates with Heterogeneous Capital Stocks

4.5 Chilean Empirical Evidence: The Peripheral and External Constraints

4.5.1 Econometric Strategy: Threshold Conditional Mixed Process Estimation

4.5.2 Empirical Estimates of the Balance-of-Payments Constraint

4.5.3 Transformation and Capacity Utilization und Financial Subordination

4.6 Comparative Analysis and Theoretical Synthesis

5 Comparative Determination: Center and Periphery

A The empirical estimations established in Section 4 demonstrate that capacity utilization cannot be understood as a universal engineering constant, an unmediated behavioral ratio, or an empirical residual. Instead, utilization is a structurally reconstructed object that emerges from the identification of the aggregate transformation elasticity θ_t, through which capitalist accumulation is converted into productive capacity.

B Crucially, the comparison between the center (the United States) and the periphery (Chile) is not a comparison between a theoretical norm and a structural deviation. Core and peripheral economies operate within the same fundamental capitalist logic of decentralized accumulation, market anarchy, and internal managerial division. What distinguishes the periphery from the center is not the presence of a different economic logic, but the distinct institutional and structural mediations through which the transformation elasticity θ_t is determined.

5.1 Structural Overaccumulation in the Center

A In the central case, capital accumulation proceeds within an advanced, self-reproducing capital-goods sector. The aggregate transformation elasticity θ_t is determined primarily by domestic class

[6] All WBOP series are available at `wbop.world` and are also hosted in the World Inequality Database (`wid.world`), together with a detailed replication package including raw data sources, methods, and codes.

distribution (ω), internal corporate financialization (ι), and the relative asset price of machinery to structures (p^{κ}).

B Under the post-war Fordist settlement (1929–1973), institutional collective bargaining and strong wage floors generated sustained distributive pressure. In accordance with the Okishio induced-innovation mechanism, cost-minimizing corporate firms responded to wage pressure by accelerating physical mechanization. Because corporate cash flows remained focused on physical plant accumulation rather than financialized asset diversion ($\iota \to 0$), distributive pressure enhanced the capacity-generating efficiency of capital, sustaining high investment dynamism and maintaining capacity utilization within a stable corridor around normal capacity ($\hat{\mu} \in [0.80, 1.05]$).

C With the collapse of Fordism and the onset of neoliberal restructuring (1974–2024), this transformation mechanism underwent a profound institutional decoupling. Real wage growth was severed from productivity gains, while corporate liquidity was increasingly diverted into intellectual property products, share buybacks, and financialized asset accumulation ($\iota \uparrow$). Simultaneously, relative machinery price inflation penalized the intensive mechanization margin ($\hat{\theta}_{\kappa p \kappa} < 0$).

D This structural shift drove the aggregate transformation elasticity permanently below unity ($\theta_t^{\text{SpecB}} < 1.00$). Bounded between 0.43 and 0.46 across the entire post-war era, $\theta_t < 1.00$ defines the **Structural Overaccumulation Regime**: a historical condition in which every 1% expansion in physical corporate capital stock yields less than a 1% expansion in potential capacity output. Under weak labor closure and financialized asset diversion, accumulation decoupling produces chronic industrial underutilization and secular stagnation.

5.2 The External Realization Trap in the Periphery

A In the peripheral case, the transformation of accumulation into productive capacity encounters a qualitatively distinct structural boundary: the absence of a domestic capital-goods sector. Industrialization under import substitution (ISI) alters the form of external dependence without eliminating it. Instead of importing finished consumer goods, the peripheral economy must import capital equipment (K^{ME}) to expand domestic productive capacity.

B Because capital-goods imports must be financed through primary commodity export earnings ($X_t \cdot ToT_t$), the foreign exchange envelope constitutes a binding macroeconomic constraint. Unlike central firms that adjust capital intensity against domestic wage bargaining, peripheral firms confront an external realization bottleneck:

C Regime A (Slack External Constraint, $D_t^{ext} = 0$). When the Prebischian Capacity to Import is unconstrained ($C_t > \hat{\gamma}$), domestic wage pressure induces cost-minimizing firms to mechanize ($\hat{\beta}_2 = 4.85^{**}$), replicating the Okishio distributive dividend observed in the core.

D Regime B (Binding Realization Trap, $D_t^{ext} = 1$). When terms-of-trade deterioration or export volume contractions depress foreign exchange capacity below the critical threshold ($C_t \leq \hat{\gamma}$), the **Peripheral Realization Penalty** activates violently ($\hat{\delta}_2 = -15.99^{***}$). The net elasticity collapses to $\hat{\beta}_2 + \hat{\delta}_2 = -11.14$.

E Under extreme external balance stress, nominal wage increases cannot translate into physical mechanization because foreign exchange scarcity blocks the importation of capital equipment. Surplus value is drained through conspicuous consumption and capital flight, causing capacity utilization to collapse. The balance of payments thus acts as the institutional channel that converts external structural constraints into internal workplace stagnation.

5.3 Synthesis: Differentiated Determinations of a Common Object

A The comparative findings for the United States and Chile demonstrate that core overaccumulation and peripheral realization traps are mutually entangled expressions of the same world-system reproduction problem. The central economy manages transformation through internal financialization and capital composition shifts, whereas the peripheral economy manages transformation through foreign exchange allocation and external balance constraints.

B The difference between center and periphery lies not in whether capacity utilization exists as an identifiable structural object, but in how the transformation of accumulation into productive capacity is structurally mediated across distinct nodes of the world capitalist economy.

6 Bounded Conclusion

A This paper has reconstructed capacity utilization as a structurally mediated object under capitalist disequilibrium. Departing from treatments that view utilization either as an exogenous engineering constant, an unmediated behavioral ratio, or an empirical regression residual, we have demonstrated that capacity utilization ($\mu_t = Y_t/Y_t^p$) becomes identifiable only downstream of the recovery of the aggregate transformation elasticity θ_t.

B By integrating Okishio's theory of decentralized market anarchy, Vidal's framework of the internally divided firm, and Marglin-style distributive grounding, we established that the transformation of capital accumulation into potential productive capacity ($g_{Y^p} = \theta_t g_K$) is endogenously conditioned by class distribution, corporate financialization, capital composition, and external balance constraints.

C Empirically, our cointegrating polynomial regression (CMPR IM-OLS) estimations delivered two major historical conclusions:

1. **Center (United States, 1929–2024):** The U.S. corporate economy operates permanently within a structural regime of capital overaccumulation ($\theta_t < 1.00$). The historical transition from post-war Fordism to neoliberal post-Fordism represents an endogenous regime shift—driven by wage suppression, relative machinery price inflation, and corporate financialization—that generated secular downward drift in capacity utilization.

1. **Periphery (Chile, 1920–2024):** In the peripheral economy, the transformation relation is non-linearly gated by foreign exchange availability. Using Chen's (2015) Profile Least Squares algorithm over the Prebischian Capacity to Import ($C_t = X_t \cdot ToT_t$), we demonstrated that when external constraints bind ($D_t^{ext} = 1$), the Okishio distributive dividend breaks down, triggering a severe peripheral realization trap ($\hat{\delta}_2 = -15.99^{***}$).

D These findings demonstrate that the difference between center and periphery is not a difference between a realized norm and a structural deviation, but a difference in how the same fundamental transformation problem is mediated under distinct world-system conditions.

E The scope of this chapter is deliberately bounded. It does not present a complete crisis theory, a fully specified model of profitability dynamics, or an exhaustive dependency framework. The chapter reconstructs capacity utilization as a structurally mediated object and identifies the transformation elasticity as the hinge of its empirical recovery across center and periphery, while leaving the broader crisis and profitability architecture for subsequent work.

References

Aglietta, M. (2000). *A Theory of Capitalist Regulation: The US Experience*. Verso.

Ahumada, J. M. (2019). *The Political Economy of Peripheral Growth: Chile in the Global Economy*. Palgrave Macmillan, Cham.

Basu, D. (2013). Replacement versus historical cost profit rates: What is the difference? when does it matter? *Metroeconomica*, 64(2):293–318.

Basu, D. (2022). *The Logic of Capital: An Introduction to Marxist Economic Theory*. Cambridge University Press, Cambridge.

Basu, D. (2026). Technical change and the rate of profit in Classical-Marxian models of economic growth. *Review of Radical Political Economics*. Forthcoming / Advance Online Publication.

Bergmann, L., Sheppard, E., and Plummer, P. S. (2009). Capitalism beyond harmonious equilibrium: Mathematics as if human agency mattered. *Environment and Planning A*, 41(2):265–283.

Blecker, R. A. and Setterfield, M. (2019). *Heterodox Macroeconomics: Models of Demand, Distribution and Growth*. Edward Elgar Publishing.

Cajas-Guijarro, J. C. (2024). An extended Goodwin model with endogenous technical change and labor supply. *Structural Change and Economic Dynamics*, 70:699–710.

Duménil, G. and Lévy, D. (1999). Being Keynesian in the short term and classical in the long term: The traverse to classical long-term equilibrium. *The Manchester School*, 67(6):684–716.

Ffrench-Davis, R. and Vivanco, D. (2016). Depreciación del capital físico, inversión neta y producto interno neto.

Foley, D. K. (1985). Say's law in marx and keynes. *Cahiers d'économie Politique / Papers in Political Economy*, pages 183–194.

Fraumeni, B. et al. (1997). The measurement of depreciation in the us national income and product accounts. *Survey of Current Business-United States Department of Commerce*, 77:7–23.

Hall, P. A. and Soskice, D. (2003). Varieties of capitalism and institutional complementarities. In *Institutional conflicts and complementarities: Monetary policy and wage bargaining institutions in EMU*, pages 43–76. Springer.

Hofman, A. A. (2000a). *The Economic Development of Latin America in the Twentieth Century*. ECLAC, Santiago.

Hofman, A. A. (2000b). Standardised capital stock estimates in Latin America: A 1950–94 update. *Cambridge Journal of Economics*, 24(1):45–86.

Julius, A. J. (2005). Steady-state growth and distribution with an endogenous direction of technical change. *Metroeconomica*, 56(1):101–125.

Kaldor, N. (1957). A model of economic growth. *The Economic Journal*, 67(268):591–624.

Kaldor, N. (1959). Problemas económicos de Chile. *El Trimestre Económico*, 26(102(2)):170–221. English original: "Economic Problems of Chile".

Kennedy, C. (1964). Induced bias in innovation and the theory of distribution. *The Economic Journal*, 74(295):541–547.

Lapavitsas, C. (2011). Theorizing financialization. *Work, Employment and Society*, 25(4):611–626.

Lewis, W. A. (1954). Economic development with unlimited supplies of labour. *The Manchester School*, 22:139–191.

Lüders, R., Díaz, J., and Wagner, G. (2016). *La República en Cifras: Historical Statistics*. Ediciones UC, Santiago.

Marx, K. (1999). *El Capital*, volume 1. Fondo de Cultura Económica, sexta re-impresión edition.

Nievas, G. and Piketty, T. (2025). Unequal exchange and North-South relations: Evidence from global trade flows and the world balance of payments 1800–2025. Working Paper 2025/11, World Inequality Lab.

Nikiforos, M. (2016). On the 'utilisation controversy': a theoretical and empirical discussion of the kaleckian model of growth and distribution. *Cambridge journal of economics*, 40(2):437–467.

Nomura, K. (2005). Duration of assets: examination of directly observed discard data in japan.

OECD (2001). Measuring capital — OECD manual: Measurement of capital stocks, consumption of fixed capital and capital services. Technical report, OECD Publishing, Paris.

Okishio, N. (1961). Technical change and the rate of profit. *Kobe University Economic Review*, 7:85–99.

Okishio, N. (2022). *The Theory of Accumulation: A Marxian Approach to the Dynamics of Capitalist Economy*. Springer Nature.

Pagano, U. (2014). The crisis of intellectual monopoly capitalism. *Cambridge Journal of Economics*, 38(6):1409–1429.

Palma, J. G. (2019). Desindustrialización, desindustrialización "prematura" y "síndrome holandés". *El Trimestre Económico*, 86(344):901–966.

Prebisch, R. (1981). *Capitalismo Periférico: Crisis y Transformación*. Fondo de Cultura Económica, México.

Pérez-Eyzaguirre, J. (2019). *Urbanización, crecimiento y cambio estructural en una economía exportadora: el caso de Chile, 1860–1940*. PhD thesis, Universidad de Chile. Tesis de doctorado.

Shaikh, A. (2016). *Capitalism: Competition, Conflict, Crises*. Oxford University Press.

Singh, M., Rathi, R., and Singh Kaswan, M. (2022). Capacity utilization in industrial sector: a structured review and implications for future research. *World Journal of Engineering*, 19(3):310–328.

Stockhammer, E. (2012). Financialization, income distribution and the crisis. *Investigación económica*, 71(279):39–70.

Sunkel, O. (1958). La inflación chilena: Un enfoque heterodoxo. *El Trimestre Económico*, 25(100(4)):570–599.

Tavani, D. (2012). Wage bargaining and induced technical change in a linear economy: Model and application to the US (1963–2003). *Structural Change and Economic Dynamics*, 23(2):117–126.

Tavani, D. and Zamparelli, L. (2018). Endogenous technical change in alternative theories of growth and distribution. In Hein, E. et al., editors, *Analytical Political Economy*, pages 139–174. Edward Elgar Publishing.

Trigg, A. B. (2020). Marx, say's law and commodity money. *Contributions to Political Economy*, 39(1):23–41.

Vernengo, M. (2006). Technology, finance, and dependency: Latin american radical political economy in retrospect. *Review of radical political economics*, 38(4):551–568.

Vidal, M. (2014). Incoherence and dysfunctionality in the institutional regulation of capitalism. *Comparative political economy of work*, pages 73–97.

Vidal, M. (2015). Fordism and the golden age of atlantic capitalism. *The Sage Handbook of the Sociology of Work and Employment. London: Sage Publications*, pages 283–305.

Vidal, M. (2019). Geriatric capitalism: Stagnation and crisis in western capitalism. In Vidal, M., Smith, T., Rotta, T., and Prew, P., editors, *The Oxford Handbook of Karl Marx*. Oxford University Press.

Vidal, M. (2022). *Management Divided: Contradictions of Labor Management*. Oxford University Press.

Vidal, M. and Peck, J. (2012). Sociological institutionalism and the socially constructed economy. In Barnes, T. J., Peck, J., and Sheppard, E., editors, *The Wiley-Blackwell Companion to Economic Geography*, pages 594–611. Wiley-Blackwell, Oxford.

Zamparelli, L. (2024a). On the positive relation between the wage share and labor productivity growth with endogenous size and direction of technical change. *Economic Modelling*, 131:106622.

Zamparelli, L. (2024b). Technical change in alternative theories of growth. Working Paper 2404, New School for Social Research, Department of Economics.

Appendix A: Proof of Proposition 3

A.1 This appendix provides the detailed derivation of the comparative statics stated in Proposition 3 (§25–26). The proof proceeds by differentiating the optimal mechanization gap (24) with respect to each state variable.

A.1 The Optimal Mechanization Gap

A.2 Recall from §3.2 that the firm maximizes the net viability surplus:

$$\max_{\kappa_t} \; c(\kappa_t) = \Phi(\kappa_t) - \kappa_t \cdot MC(\omega_{t-1}, p_t^{\kappa}) \tag{73}$$

where $\Phi(\kappa_t) = \lambda_0 + \lambda_1 \kappa_t + \lambda_2 \kappa_t^2$ is the mechanization frontier (19), and $MC(\omega_{t-1}, p_t^{\kappa}) = \gamma_0 - \gamma_1 \omega_{t-1} + \gamma_2 p_t^{\kappa}$ is the marginal viability cost (21). Substituting and differentiating:

$$c(\kappa_t) = \lambda_0 + [\lambda_1 - \gamma_0 + \gamma_1 \omega_{t-1} - \gamma_2 p_t^{\kappa}]\, \kappa_t + \lambda_2 \kappa_t^2 \tag{74}$$

$$\frac{dc}{d\kappa_t} = \lambda_1 - \gamma_0 + \gamma_1 \omega_{t-1} - \gamma_2 p_t^{\kappa} + 2\lambda_2 \kappa_t^* = 0 \tag{75}$$

Solving (75) under Assumption 1 ($\lambda_2 < 0$) yields the optimal mechanization gap:

$$\kappa_t^*(\omega_{t-1}, p_t^{\kappa}) = \frac{(\gamma_0 - \lambda_1) - \gamma_1 \omega_{t-1} + \gamma_2 p_t^{\kappa}}{2\lambda_2} \tag{76}$$

which reproduces (24).

A.2 The Distributive Channel

A.3 **Proof of** (25). Differentiating (76) with respect to ω_{t-1}:

$$\frac{\partial \kappa_t^*}{\partial \omega_{t-1}} = \frac{-\gamma_1}{2\lambda_2} \tag{77}$$

Under Assumption 1, $\lambda_2 < 0$. By Assumption 3 ($\gamma_1 > 0$), the numerator $-\gamma_1 < 0$ and the denominator $2\lambda_2 < 0$. Therefore:

$$\frac{\partial \kappa_t^*}{\partial \omega_{t-1}} = \frac{-\gamma_1}{2\lambda_2} > 0 \tag{78}$$

A rising wage share forces cost-minimizing firms to widen the mechanization gap, substituting machinery for labor. ■

A.3 The Relative Price Channel

A.4 **Proof of** (26). Differentiating (76) with respect to p_t^{κ}:

$$\frac{\partial \kappa_t^*}{\partial p_t^{\kappa}} = \frac{\gamma_2}{2\lambda_2} \tag{79}$$

Under Assumption 1, $\lambda_2 < 0$. By Assumption 3 ($\gamma_2 > 0$), the numerator $\gamma_2 > 0$ and the denominator $2\lambda_2 < 0$. Therefore:

$$\frac{\partial \kappa_t^*}{\partial p_t^\kappa} = \frac{\gamma_2}{2\lambda_2} < 0 \tag{80}$$

A rising relative price of machinery suppresses the optimal mechanization gap, as the structural cost of the composition shift penalizes mechanization. ■

A.4 The Cost Hurdle Formulation (Alternative Derivation)

A.5 An equivalent derivation expresses the FOC as a cost hurdle. Define the effective cost hurdle of mechanization:

$$H(\omega_{t-1}, p_t^\kappa) \equiv \frac{MC(\omega_{t-1}, p_t^\kappa)}{1} = \gamma_0 - \gamma_1 \omega_{t-1} + \gamma_2 p_t^\kappa \tag{81}$$

The FOC (75) can be rewritten as:

$$\Phi'(\kappa_t^*) = H(\omega_{t-1}, p_t^\kappa) \tag{82}$$

The firm mechanizes up to the point where the marginal productivity of the mechanization gap equals the effective cost hurdle. The comparative statics follow from the implicit function theorem applied to (82):

$$\frac{\partial \kappa_t^*}{\partial \omega_{t-1}} = -\frac{\partial H / \partial \omega_{t-1}}{\Phi''(\kappa_t^*)} = -\frac{-\gamma_1}{2\lambda_2} > 0 \tag{83}$$

$$\frac{\partial \kappa_t^*}{\partial p_t^\kappa} = -\frac{\partial H / \partial p_t^\kappa}{\Phi''(\kappa_t^*)} = -\frac{\gamma_2}{2\lambda_2} < 0 \tag{84}$$

Both derivations yield identical results, confirming Proposition 3.

A Data Construction and Measurement Provenance

A The main text of §4.1 keeps only what identification requires: the constructed capital stocks, the distinction between depreciation and retirement, the analytical roles of gross and net capital, and the rule that cross-country comparability rests on accounting consistency rather than on harmonized source vintages. Everything needed to reproduce those constructions, but not to follow the argument, is collected here.

A.1 The generalized perpetual inventory method

A [**PLACEHOLDER**] Full statement of the GPIM construction: survival function and its parameterization, service-life assumptions by asset class, the initialization of the capital stock at the start of each series, and the treatment of the discard rate MATHPLACEHOLDER0XYZ on gross capital in operation as distinct from the net depreciation rate MATHPLACEHOLDER1XYZ. Source material sits in the archived maintenance audits at `reports/maintenance/` — specifically the gross-survival remediation, input-initialization, price-deflator provenance, and source-price-seed audits.

A.2 Price bases and deflator provenance

A [**PLACEHOLDER**] Deflator chain for each series and the base-year conventions. The Chilean capital stock and GDP series share a 2003 CLP base and require no rebasing, which is the reason the Chilean accounting closes without a splice at that seam; this needs stating explicitly with its source. Pre-1940 Chilean relative capital-asset prices are proxied by a US wholesale price index, and the consequences of that proxy for the price channel need to be stated here rather than left in a table footnote.

A.3 Source reconciliation and splice diagnostics

A [**PLACEHOLDER**] Series-by-series reconciliation against source vintages, splice points and the diagnostics run at each, and the repository provenance of every input. Source material is in the archived `reports/US_Wage_Share_Data_Provenance_Note/99_archive/` and the corresponding Chilean lane.

A.4 The wage-share measures

A Four wage-share measures enter the estimation and the distinctions between them do analytical work, so they are set out here in full rather than by reference.

B The baseline measure takes compensation over value added for the non-financial corporate sector. The gross and net variants differ in whether consumption of fixed capital sits in the denominator; because that quantity rises mechanically with capital intensity, the two variants respond differently to exactly the mechanization the interaction term is meant to identify, which is why §4.2 treats the gross-basis failure as evidence rather than as noise.

C The two Shaikh-Tonak variants further separate productive from supervisory labour compensation. Tier 1 isolates variable capital on a first pass; Tier 2 applies the supervisory correction in full. The theoretical motivation is that cost-minimizing choice of technique responds to shop-floor labour cost, not to an aggregate that folds managerial overhead into the same ratio.

D [**PLACEHOLDER**] Construction detail for each of the four measures — the exact NIPA line items, the supervisory split and its source, and the crosswalk between the four — belongs in this subsection.

A.5 The level anchor

A Capacity utilization is a ratio whose numerator is observed and whose denominator is estimated. Estimation recovers the transformation relation and therefore the shape of the productive-capacity path, but not its level: any scalar multiple of $\hat{Y}_t^p$ fits equally well. The level is fixed by anchoring utilization to unity in a benchmark year chosen to approximate frontier utilization.

B This normalization anchors the series. It does not define the object. The distinction matters because it is precisely what separates this construction from approaches that recover utilization as a residual: here the transformation elasticity is identified first and the level is imposed afterwards, so the anchor year carries no weight in identifying θ.

C [**PLACEHOLDER**] Statement of the pinch-year selection rule, the chosen benchmark year for each case with its justification, and sensitivity of the reconstructed level path to alternative anchor years. Protocol is at `chapter2_vault/04_data_measurement/D03_capacity_utilization_level_anchor_pinch_year_protocol.md`.

B The Specification Space and Model Selection

A The main text reports one preferred specification per case. This appendix reports the space those specifications were selected from, which alternatives were estimated, and why each is not carried forward. Presenting a single preferred model without its search space would leave the reader unable to judge how much the result depends on choices made along the way.

B.1 The combinatorial grid

A Specification A varies three binary dimensions — wage-share denominator (gross versus net value added), interaction form (standard versus effective, the latter scaled by the financialization retention index $(1-\iota_{t-1})$), and the presence of Bai-Perron step level-shifts — across three wage-share measures, giving the estimation matrix reported in §4.2. Specification B adds a fourth dimension by disaggregating capital into structures k^{NRC} and the mechanization gap κ_t, and by admitting a relative-price interaction alongside the distributive one.

B Every cell of these grids was estimated. None was dropped for producing an inconvenient result.

B.2 Why alternatives are not carried forward

A Specifications fall out of the preferred set for reasons of different weight, and the distinction matters for interpretation.

B Some are excluded on **admissibility**: a regressor fails the integration-order gate as actually constructed, including its lag structure, or a generated interaction inherits $I(2)$ behaviour from its components. Theoretical motivation does not rescue these. The theory describes a decision mechanism; the stochastic evidence determines whether a given observable proxy for that mechanism can enter a long-run regression at all.

C Some are excluded on **measurement**: the gross-value-added wage shares are admissible but uninformative, because consumption of fixed capital rises mechanically with capital intensity and so contaminates the distributive signal precisely along the dimension the interaction is meant to identify. Their failure is itself a result, not a nuisance, and §4.2 reads it as one.

D Some are excluded on **redundancy**: the step-dummy variants sustain cointegration but add nothing once the distributive-financial interaction is specified, which is the basis for carrying forward the no-break specification.

E None of the estimated specifications was excluded because it failed to cointegrate.

F [**PLACEHOLDER**] The full consolidated grid — the MATHPLACEHOLDER0XYZ Specification A and MATHPLACEHOLDER1XYZ Specification B permutations under initial-state centering, with each cell's disposition and the economic reason for it — is already written at `chapter2_vault/06_paper_facing/results/99_legacy/S05_Appendix.md` (status: locked). It needs transposing into this appendix with its internal governance vocabulary translated into plain econometric language: the vault's four-way ledger of dispositions is an internal audit device and must not surface in paper-facing text.

G [**PLACEHOLDER**] The integration-order verdict table — each candidate regressor mapped to its theoretical basis, its MATHPLACEHOLDER0XYZ evidence on both levels and first differences from at least two complementary tests, and its final disposition — belongs here. Source material is in the archived `reports/US_Integration_Order_Appendix/99_archive/`.

B.3 Counterfactual restrictions

A Two restrictions test whether the estimated transformation elasticity is doing work the data support rather than work the specification imposes.

B The first restricts $\theta = 1$, the Harrodian knife-edge, and asks what the capacity series and the implied utilization path look like under that restriction. This is the case in which accumulation converts one-for-one into productive capacity, and it is the implicit assumption of any filtering approach that fits output to capital without separately identifying the transformation relation.

C The second removes the step dummies entirely and re-estimates, to establish how much of the reported structure survives without any exogenous break structure.

D [**PLACEHOLDER**] Both counterfactuals are specified but not yet reported. Each requires a re-estimation pass and a comparison table against the preferred specification.

C Cointegration Diagnostics and the Monte Carlo Critical-Value Protocol

A The main text reports one cointegration statistic per specification: the Vogelsang and Wagner (2014) Type 2 test. This appendix reports the full diagnostic set behind that verdict, states the critical values, and explains why the Type 2 test rather than the residual tests carries the reading.

C.1 Why the Type 2 test governs

A The three tests in use do not stand on equal footing, and treating them as a co-equal triad obscures what each is doing.

B The residual Engle-Granger ADF test asks whether the estimated residual series contains a unit root. Its null is *no* cointegration, so rejection supports the long-run relation. The Shin LM test inverts this: its null is cointegration, and rejection indicates spurious regression. The Vogelsang-Wagner Type 2 test also takes cointegration as its null, but it differs from both residual tests in a way that matters for this chapter. Rather than testing a property of a residual series

constructed after the fact, it tests a restriction on the IM-OLS estimator itself — that the auxiliary coefficient vector λ_2 is zero, which holds under genuine cointegration and fails under spurious regression. Because the specifications here carry generated interaction regressors that have been FWL-orthogonalized against the base capital terms and the deterministic space, a residual-based test inherits whatever the orthogonalization did to the residual's serial dependence. The Type 2 statistic does not.

C That is the substantive reason for the hierarchy, and it is the one the estimation protocol adopts: the Type 2 test is primary and binding, the residual ADF and Shin LM tests are secondary robustness checks, and Johansen and Phillips-Ouliaris statistics are background diagnostics that do not enter the paper.

C.2 Reading the stars: opposed nulls

A Because the nulls run in opposite directions, an identical typographic convention carries opposite meanings across rows of the tables below. This is worth stating plainly rather than leaving to a footnote.

B For the residual ADF test, stars indicate rejection of no cointegration, and are therefore the favourable result. For the Shin LM and Vogelsang-Wagner Type 2 tests, stars indicate rejection of cointegration, and are therefore the unfavourable result. A well-behaved specification shows stars on its ADF row and none on the other two.

C.3 Critical values

Table 4: Critical Values for the Cointegration Diagnostics

Test	Source	10%	5%	1%
Residual ADF (t_{ADF})	MacKinnon (2010), Case 2, constant only, $m = 4$	-3.81	-4.10	-4.64
Shin LM (C_{Shin}), $k = 2$	Shin (1994), Table 1, demeaned C_μ	0.163	0.221	0.380
Shin LM (C_{Shin}), $k = 3$	Shin (1994), Table 1, demeaned C_μ	0.121	0.159	0.271
VW Type 2 ($S_{\mathrm{Type\ 2}}$)	Vogelsang and Wagner (2014), fixed-b, Bartlett kernel, intercept-only	jointly indexed by (b, k)	no single constant applies	see D.3 note

A Here m counts the total number of $I(1)$ variables in the system, one dependent variable plus the stochastic regressors, with step dummies excluded from the count; k counts stochastic $I(1)$ regressors alone. Exogenous $I(0)$ controls sit in the deterministic matrix and so do not enter k.

B The last row is the substantive point. Vogelsang-Wagner fixed-b critical values are not constants. They vary by roughly an order of magnitude across the tabulated bandwidth-ratio range and depend on the regressor count, so every model must be evaluated against the cell matching its own (b, k) pair. Each table below reports the bandwidth ratio b for exactly this reason.

C

> [**PLACEHOLDER**] The Type 2 rows in the tables below carry no significance stars, because the statistics have not yet been evaluated against the MATHPLACEHOLDER0XYZ-indexed fixed-MATHPLACEHOLDER1XYZ table digitized in `codes/COMMON_VW_fixedb_critical_values.R`. Earlier drafts compared them against a single hard-coded triple (MATHPLACEHOLDER2XYZ / MATHPLACEHOLDER3XYZ / MATHPLACEHOLDER4XYZ) applied uniformly across models regardless of bandwidth or regressor count; a 2026-08-03 audit established that this triple matches no cell of the published table, and those comparisons have been withdrawn rather than reproduced here. Re-running the estimation against the digitized module is the outstanding work. Until it is done, this appendix reports the statistics and the critical-value regime, but not a starred verdict.

C.4 Constructing the residual statistics

A Two implementation details determine whether the residual diagnostics measure what they are supposed to measure.

B The Shin LM statistic is evaluated on the IM-OLS partial sum residuals $U_t = \sum e_t$ directly, with no further accumulation. Because U_t is already an integrated partial sum by construction, applying a second cumulative sum produces an $I(2)$ series and inflates the statistic. The error is easy to make and hard to see, since the inflated statistic remains finite and ordered plausibly across models.

C The regressors entering the residual tests are the deterministically detrended series — the residuals of $\mathbf{x}$ regressed on the step dummies and their partial sums — from which the reverse cumulative sum regressors are then built. Skipping the detrending leaves deterministic trend contamination in the test.

C.5 Small-sample and lag-augmentation caveats

A Two further constraints bear on how the residual diagnostics below should be read.

B Residual unit-root tests must be lag-augmented with an information-criterion-selected lag order. A plain Dickey-Fuller regression with zero lagged differences is not admissible where the residuals are serially correlated, which in this setting means everywhere. Where a sub-sample is short — the Fordist window gives $T = 44$ — small-sample bias pushes the true critical values more negative than the asymptotic boundaries tabulated above, so marginal significance near -4.10 in a sub-sample should not be read as a clean rejection.

C Second, step dummies test only for level shifts. They cannot absorb a shift in the cointegrating vector itself, so a specification that answers detected breaks with step dummies alone has not addressed slope instability and is not described as break-robust anywhere in this chapter. Appendix E takes up parameter stability directly.

C.6 Specification A: aggregate capital

Table 5: Specification Set 1: Standard Wage Interaction (GVA Basis) — Cointegration Diagnostics and Collinearity

Dependent Variable: $y_t^{\mathbf{NFC}}$ (Log NFC Real NVA)								
Model	M1.1a	M1.1b	M1.2a	M1.2b	M1.3a	M1.3b	M1.4a	M1.4b
Collinearity Diagnostics (VIF)								
VIF (k_{cap})	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
VIF (Interaction)	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Cointegration Test Triad								
Resid. ADF (t_{ADF}) [H_0: No Coint.]	-2.41	-2.81	-2.33	-2.73	-4.15**	-3.56*	-3.40	-3.55*
Shin LM (C_{Shin}) [H_0: Coint.]	0.06	0.04	0.06	0.04	0.05	0.03	0.03	0.03
VW Type 2 ($S_{\text{Type 2}}$) [H_0: Coint.]	3.30	11.34	2.70	11.41	5.82	9.68	1.46	9.08
Bandwidth Ratio (b)	0.2746	0.1872	0.2864	0.1934	0.2852	0.1997	0.2665	0.2731
Sample Size (T)	94	94	94	94	94	94	94	94

Note: Full cointegration diagnostics for the estimates reported in the main text. **Star reversal warning:** for the Residual ADF test, significance stars indicate rejection of No Cointegration (supporting cointegration). For the Shin LM test, significance stars indicate rejection of Cointegration (indicating spurious regression). The Vogelsang–Wagner Type 2 test shares the Shin null (H_0: cointegration) and its stars read the same way. Critical values: Residual ADF (MacKinnon 2010, Case 2, constant only, $m = 4$) -3.81 / -4.10 / -4.64 at 10% / 5% / 1%; Shin LM (Shin 1994, Table 1, demeaned C_μ) 0.163 / 0.221 / 0.380 for $k = 2$ and 0.121 / 0.159 / 0.271 for $k = 3$, where k counts stochastic $I(1)$ regressors; Vogelsang–Wagner Type 2 critical values are fixed-b and jointly indexed by (b, k), so no single constant applies across models [**PENDING: see the note to the corresponding main-text table**]. Bandwidth ratio b is the model's HAC bandwidth divided by T.

Table 6: Specification Set 1: Standard Wage Interaction (NVA Basis) — Cointegration Diagnostics and Collinearity

Dependent Variable: $y_t^{\mathbf{NFC}}$ (Log NFC Real NVA)								
Model	M1.5a	M1.5b	M1.6a	M1.6b	M1.7a	M1.7b	M1.8a	M1.8b
Collinearity Diagnostics (VIF)								
VIF (k_{cap})	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
VIF (Interaction)	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Cointegration Test Triad								
Resid. ADF (t_{ADF}) [H_0: No Coint.]	-2.41	-2.81	-2.41	-2.81	-2.41	-2.81	-2.94	-2.95
Shin LM (C_{Shin}) [H_0: Coint.]	0.06	0.04	0.06	0.04	0.06	0.04	0.04	0.03
VW Type 2 ($S_{\text{Type 2}}$) [H_0: Coint.]	3.30	11.34	3.30	11.34	3.30	11.34	0.05	11.94
Bandwidth Ratio (b)	0.2746	0.1872	0.2746	0.1872	0.2746	0.1872	0.1940	0.1935
Sample Size (T)	94	94	94	94	94	94	94	94

Note: Full cointegration diagnostics for the estimates reported in the main text. **Star reversal warning:** for the Residual ADF test, significance stars indicate rejection of No Cointegration (supporting cointegration). For the Shin LM test, significance stars indicate rejection of Cointegration (indicating spurious regression). The Vogelsang–Wagner Type 2 test shares the Shin null (H_0: cointegration) and its stars read the same way. Critical values: Residual ADF (MacKinnon 2010, Case 2, constant only, $m = 4$) -3.81 / -4.10 / -4.64 at 10% / 5% / 1%; Shin LM (Shin 1994, Table 1, demeaned C_μ) 0.163 / 0.221 / 0.380 for $k = 2$ and 0.121 / 0.159 / 0.271 for $k = 3$, where k counts stochastic $I(1)$ regressors; Vogelsang–Wagner Type 2 critical values are fixed-b and jointly indexed by (b, k), so no single constant applies across models [**PENDING: see the note to the corresponding main-text table**]. Bandwidth ratio b is the model's HAC bandwidth divided by T.

Table 7: Specification Set 2: Effective Wage Interaction (GVA Basis) — Cointegration Diagnostics and Collinearity

Dependent Variable: $y_t^{\mathbf{NFC}}$ (Log NFC Real NVA)								
Model	M2.1a	M2.1b	M2.2a	M2.2b	M2.3a	M2.3b	M2.4a	M2.4b
Collinearity Diagnostics (VIF)								
VIF (k_{cap})	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
VIF (Interaction)	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Cointegration Test Triad								
Resid. ADF (t_{ADF}) [H_0: No Coint.]	-2.32	-3.06	-2.26	-2.96	-4.02**	-2.93	-3.27	-2.90
Shin LM (C_{Shin}) [H_0: Coint.]	0.06	0.04	0.06	0.04	0.05	0.03	0.03	0.03
VW Type 2 ($S_{\text{Type 2}}$) [H_0: Coint.]	10.76	9.07	6.17	9.61	4.39	7.88	0.73	4.55
Bandwidth Ratio (b)	0.2976	0.1687	0.3066	0.1732	0.2908	0.1765	0.2713	0.2047
Sample Size (T)	94	94	94	94	94	94	94	94

Note: Full cointegration diagnostics for the estimates reported in the main text. **Star reversal warning:** for the Residual ADF test, significance stars indicate rejection of No Cointegration (supporting cointegration). For the Shin LM test, significance stars indicate rejection of Cointegration (indicating spurious regression). The Vogelsang–Wagner Type 2 test shares the Shin null (H_0: cointegration) and its stars read the same way. Critical values: Residual ADF (MacKinnon 2010, Case 2, constant only, $m = 4$) -3.81 / -4.10 / -4.64 at 10% / 5% / 1%; Shin LM (Shin 1994, Table 1, demeaned C_μ) 0.163 / 0.221 / 0.380 for $k = 2$ and 0.121 / 0.159 / 0.271 for $k = 3$, where k counts stochastic $I(1)$ regressors; Vogelsang–Wagner Type 2 critical values are fixed-b and jointly indexed by (b, k), so no single constant applies across models [**PENDING: see the note to the corresponding main-text table**]. Bandwidth ratio b is the model's HAC bandwidth divided by T.

Table 8: Specification Set 2: Effective Wage Interaction (NVA Basis) — Cointegration Diagnostics and Collinearity

Dependent Variable: $y_t^{\mathbf{NFC}}$ (Log NFC Real NVA)								
Model	M2.5a	M2.5b	M2.6a	M2.6b	M2.7a	M2.7b	M2.8a	M2.8b
Collinearity Diagnostics (VIF)								
VIF (k_{cap})	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
VIF (Interaction)	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Cointegration Test Triad								
Resid. ADF (t_{ADF}) [H_0: No Coint.]	-2.32	-3.06	-2.32	-3.06	-2.32	-3.06	-2.96	-3.82**
Shin LM (C_{Shin}) [H_0: Coint.]	0.06	0.04	0.06	0.04	0.06	0.04	0.04	0.03
VW Type 2 ($S_{\text{Type 2}}$) [H_0: Coint.]	10.76	9.07	10.76	9.07	10.76	9.07	0.46	13.69
Bandwidth Ratio (b)	0.2976	0.1687	0.2976	0.1687	0.2976	0.1687	0.1873	0.1651
Sample Size (T)	94	94	94	94	94	94	94	94

Note: Full cointegration diagnostics for the estimates reported in the main text. **Star reversal warning:** for the Residual ADF test, significance stars indicate rejection of No Cointegration (supporting cointegration). For the Shin LM test, significance stars indicate rejection of Cointegration (indicating spurious regression). The Vogelsang–Wagner Type 2 test shares the Shin null (H_0: cointegration) and its stars read the same way. Critical values: Residual ADF (MacKinnon 2010, Case 2, constant only, $m = 4$) -3.81 / -4.10 / -4.64 at 10% / 5% / 1%; Shin LM (Shin 1994, Table 1, demeaned C_μ) 0.163 / 0.221 / 0.380 for $k = 2$ and 0.121 / 0.159 / 0.271 for $k = 3$, where k counts stochastic $I(1)$ regressors; Vogelsang–Wagner Type 2 critical values are fixed-b and jointly indexed by (b, k), so no single constant applies across models [**PENDING: see the note to the corresponding main-text table**]. Bandwidth ratio b is the model's HAC bandwidth divided by T.

C.7 Specification B: heterogeneous capital

Table 9: Specification Set 1: Standard Wage Interaction (GVA Basis) — Cointegration Diagnostics and Collinearity

Dependent Variable: $y_t^{\mathbf{NFC}}$ (Log NFC Real GVA)						
Model	M2.1a	M2.2b	M2.5a	M2.6b	M2.9a	M2.10b
Collinearity Diagnostics (VIF)						
VIF (k^{NRC})	1.69	1.69	1.69	1.69	1.69	1.69
VIF (κ)	1.69	1.69	1.69	1.69	1.69	1.69
VIF (Wage Interaction)	1.32	1.06	1.15	1.01	1.01	1.01
VIF (Price Interaction)	1.32	1.06	1.15	1.01	1.01	1.01
Cointegration Test Triad						
Resid. ADF (t_{ADF}) [H_0: No Coint.]	-4.16*	-4.46**	-4.26*	-4.69**	-4.45**	-4.80**
Shin LM (C_{Shin}) [H_0: Coint.]	0.05	0.03	0.05	0.03	0.04	0.02
VW Type 2 ($S_{\mathrm{Type\ 2}}$) [H_0: Coint.]	16.11	9.55	16.25	11.41	16.74	7.66
Bandwidth Ratio (b)	0.2088	0.1758	0.2114	0.1744	0.2088	0.1759
Sample Size (T)	94	94	94	94	94	94

Note: Full cointegration diagnostics for the estimates reported in the main text. **Star reversal warning:** for the Residual ADF test, significance stars indicate rejection of No Cointegration (supporting cointegration). For the Shin LM test, significance stars indicate rejection of Cointegration (indicating spurious regression). The Vogelsang–Wagner Type 2 test shares the Shin null (H_0: cointegration) and its stars read the same way. Critical values: Residual ADF (MacKinnon 2010, Case 2, constant only, $m = 4$) -3.81 / -4.10 / -4.64 at 10% / 5% / 1%; Shin LM (Shin 1994, Table 1, demeaned C_μ) 0.163 / 0.221 / 0.380 for $k = 2$ and 0.121 / 0.159 / 0.271 for $k = 3$, where k counts stochastic $I(1)$ regressors; Vogelsang–Wagner Type 2 critical values are fixed-b and jointly indexed by (b, k), so no single constant applies across models [**PENDING: see the note to the corresponding main-text table**]. Bandwidth ratio b is the model's HAC bandwidth divided by T.

Table 10: Specification Set 1: Standard Wage Interaction (NVA Basis) — Cointegration Diagnostics and Collinearity

Dependent Variable: $y_t^{\mathbf{NFC}}$ (Log NFC Real GVA)						
Model	M2.3a	M2.4b	M2.7a	M2.8b	M2.11a	M2.12b
Collinearity Diagnostics (VIF)						
VIF (k^{NRC})	1.69	1.69	1.69	1.69	1.69	1.69
VIF (κ)	1.69	1.69	1.69	1.69	1.69	1.69
VIF (Wage Interaction)	1.32	1.06	1.32	1.06	1.07	1.02
VIF (Price Interaction)	1.32	1.06	1.32	1.06	1.07	1.02
Cointegration Test Triad						
Resid. ADF (t_{ADF}) [H_0: No Coint.]	-4.16*	-4.46**	-4.16*	-4.46**	-4.42**	-4.79**
Shin LM (C_{Shin}) [H_0: Coint.]	0.05	0.03	0.05	0.03	0.04	0.02
VW Type 2 ($S_{\mathrm{Type\ 2}}$) [H_0: Coint.]	16.11	9.55	16.11	9.55	16.40	8.61
Bandwidth Ratio (b)	0.2088	0.1758	0.2088	0.1758	0.2133	0.1758
Sample Size (T)	94	94	94	94	94	94

Note: Full cointegration diagnostics for the estimates reported in the main text. **Star reversal warning:** for the Residual ADF test, significance stars indicate rejection of No Cointegration (supporting cointegration). For the Shin LM test, significance stars indicate rejection of Cointegration (indicating spurious regression). The Vogelsang–Wagner Type 2 test shares the Shin null (H_0: cointegration) and its stars read the same way. Critical values: Residual ADF (MacKinnon 2010, Case 2, constant only, $m = 4$) -3.81 / -4.10 / -4.64 at 10% / 5% / 1%; Shin LM (Shin 1994, Table 1, demeaned C_μ) 0.163 / 0.221 / 0.380 for $k = 2$ and 0.121 / 0.159 / 0.271 for $k = 3$, where k counts stochastic $I(1)$ regressors; Vogelsang–Wagner Type 2 critical values are fixed-b and jointly indexed by (b, k), so no single constant applies across models [**PENDING: see the note to the corresponding main-text table**]. Bandwidth ratio b is the model's HAC bandwidth divided by T.

Table 11: Specification Set 2: Effective Wage Interaction (GVA Basis) — Cointegration Diagnostics and Collinearity

Dependent Variable: $y_t^{\mathbf{NFC}}$ (Log NFC Real GVA)						
Model	M2.13a	M2.14b	M2.17a	M2.18b	M2.21a	M2.22b
Collinearity Diagnostics (VIF)						
VIF (k^{NRC})	1.69	1.69	1.69	1.69	1.69	1.69
VIF (κ)	1.69	1.69	1.69	1.69	1.69	1.69
VIF (Wage Interaction)	1.18	1.03	1.03	1.00	1.00	1.00
VIF (Price Interaction)	1.18	1.03	1.03	1.00	1.00	1.00
Cointegration Test Triad						
Resid. ADF (t_{ADF}) [H_0: No Coint.]	-4.33*	-4.58**	-4.40*	-4.77**	-4.44**	-4.88**
Shin LM (C_{Shin}) [H_0: Coint.]	0.03	0.02	0.03	0.02	0.05	0.02
VW Type 2 ($S_{\mathrm{Type\ 2}}$) [H_0: Coint.]	14.97	5.93	14.58	3.71	16.05	3.21
Bandwidth Ratio (b)	0.1854	0.1731	0.1921	0.1735	0.2125	0.1646
Sample Size (T)	94	94	94	94	94	94

Note: Full cointegration diagnostics for the estimates reported in the main text. **Star reversal warning:** for the Residual ADF test, significance stars indicate rejection of No Cointegration (supporting cointegration). For the Shin LM test, significance stars indicate rejection of Cointegration (indicating spurious regression). The Vogelsang–Wagner Type 2 test shares the Shin null (H_0: cointegration) and its stars read the same way. Critical values: Residual ADF (MacKinnon 2010, Case 2, constant only, $m = 4$) -3.81 / -4.10 / -4.64 at 10% / 5% / 1%; Shin LM (Shin 1994, Table 1, demeaned C_μ) 0.163 / 0.221 / 0.380 for $k = 2$ and 0.121 / 0.159 / 0.271 for $k = 3$, where k counts stochastic $I(1)$ regressors; Vogelsang–Wagner Type 2 critical values are fixed-b and jointly indexed by (b, k), so no single constant applies across models [**PENDING: see the note to the corresponding main-text table**]. Bandwidth ratio b is the model's HAC bandwidth divided by T.

Table 12: Specification Set 2: Effective Wage Interaction (NVA Basis) — Cointegration Diagnostics and Collinearity

Dependent Variable: $y_t^{\mathbf{NFC}}$ (Log NFC Real GVA)						
Model	M2.15a	M2.16b	M2.19a	M2.20b	M2.23a	M2.24b
Collinearity Diagnostics (VIF)						
VIF (k^{NRC})	1.69	1.69	1.69	1.69	1.69	1.69
VIF (κ)	1.69	1.69	1.69	1.69	1.69	1.69
VIF (Wage Interaction)	1.18	1.03	1.18	1.03	1.02	1.00
VIF (Price Interaction)	1.18	1.03	1.18	1.03	1.02	1.00
Cointegration Test Triad						
Resid. ADF (t_{ADF}) [H_0: No Coint.]	-4.33*	-4.58**	-4.33*	-4.58**	-4.44**	-4.86**
Shin LM (C_{Shin}) [H_0: Coint.]	0.03	0.02	0.03	0.02	0.05	0.02
VW Type 2 ($S_{\mathrm{Type\ 2}}$) [H_0: Coint.]	14.97	5.93	14.97	5.93	15.72	4.59
Bandwidth Ratio (b)	0.1854	0.1731	0.1854	0.1731	0.2137	0.1661
Sample Size (T)	94	94	94	94	94	94

Note: Full cointegration diagnostics for the estimates reported in the main text. **Star reversal warning:** for the Residual ADF test, significance stars indicate rejection of No Cointegration (supporting cointegration). For the Shin LM test, significance stars indicate rejection of Cointegration (indicating spurious regression). The Vogelsang–Wagner Type 2 test shares the Shin null (H_0: cointegration) and its stars read the same way. Critical values: Residual ADF (MacKinnon 2010, Case 2, constant only, $m = 4$) -3.81 / -4.10 / -4.64 at 10% / 5% / 1%; Shin LM (Shin 1994, Table 1, demeaned C_μ) 0.163 / 0.221 / 0.380 for $k = 2$ and 0.121 / 0.159 / 0.271 for $k = 3$, where k counts stochastic $I(1)$ regressors; Vogelsang–Wagner Type 2 critical values are fixed-b and jointly indexed by (b, k), so no single constant applies across models [**PENDING: see the note to the corresponding main-text table**]. Bandwidth ratio b is the model's HAC bandwidth divided by T.

C.8 Chile: threshold specification

Table 13: Chile Peripheral Extension: Threshold CMPR IM-OLS Cointegration Estimates (1920–2024) — Cointegration Diagnostics and Collinearity

Dependent Variable: y_t (Log Aggregate Chilean Real GDP)					
Parameter / Variable	Spec M3.1a (Baseline)	Spec M3.2a (Mediated)	Spec M3.3a (T-CMPR Slack E_A)	Spec M3.3b (T-CMPR Trap E_B)	Spec M3.4a (Full T-CMPR)
Collinearity Diagnostics (VIF)					
VIF (k^{NRC})	1.42	1.42	1.42	1.42	1.42
VIF (κ)	1.42	1.42	1.42	1.42	1.42
VIF (Wage / Threshold Interactions)	1.02	1.05	1.02	1.04	1.04
Cointegration Test Triad					
Resid. ADF (t_{ADF}) [H_0: No Coint.]	-4.12**	-4.35**	-4.40**	-4.18**	-4.62***
Shin LM (C_{Shin}) [H_0: Coint.]	0.04	0.03	0.03	0.04	0.02
VW Type 2 ($S_{\mathrm{Type\ 2}}$) [H_0: Coint.]	9.15	7.82	7.25	11.40	6.85
Sample Size (T)	104	104	104	104	104

Note: Full cointegration diagnostics for the estimates reported in the main text. **Star reversal warning:** for the Residual ADF test, significance stars indicate rejection of No Cointegration (supporting cointegration). For the Shin LM test, significance stars indicate rejection of Cointegration (indicating spurious regression). The Vogelsang–Wagner Type 2 test shares the Shin null (H_0: cointegration) and its stars read the same way. Critical values: Residual ADF (MacKinnon 2010, Case 2, constant only, $m = 4$) -3.81 / -4.10 / -4.64 at 10% / 5% / 1%; Shin LM (Shin 1994, Table 1, demeaned C_μ) 0.163 / 0.221 / 0.380 for $k = 2$ and 0.121 / 0.159 / 0.271 for $k = 3$, where k counts stochastic $I(1)$ regressors; Vogelsang–Wagner Type 2 critical values are fixed-b and jointly indexed by (b, k), so no single constant applies across models [**PENDING: see the note to the corresponding main-text table**]. Bandwidth ratio b is the model's HAC bandwidth divided by T.

C.9 The Monte Carlo protocol

A Coefficient-level significance throughout the chapter comes from a 100,000-pass Monte Carlo simulation that reproduces the exact data-generating process of each specification variant, including whether step level-shifts are present. This is not a convenience. Fixed-b critical values for a single-coefficient t-test are systematically larger in absolute value than the standard-normal cutoffs of 1.645, 1.960, and 2.576, so applying the normal cutoffs to an estimator whose stated methodology is fixed-b HAR overstates significance. Re-auditing Specification A under the corrected procedure moved 39 coefficient-level stars, every one of them a downgrade.

B For the Chilean threshold specification the simulation additionally replicates the Profile Least Squares grid search over the capacity-to-import threshold inside each pass. Without this the inference would be conditional on a threshold treated as known when it was in fact estimated from the same data, and the grid search consumes degrees of freedom that the asymptotic bounds do not account for.

C [**PLACEHOLDER**] Tail-convergence evidence for the simulation — the MATHPLACEHOLDER0XYZ, MATHPLACEHOLDER1XYZ, and MATHPLACEHOLDER2XYZ convergence diagnostics at MATHPLACEHOLDER3XYZ, and the inflation of the 95th-percentile critical values under structural breaks — exists at `chapter2_vault/06_paper_facing/results/03_monte_carlo_tail_convergence_report.md` with four supporting figures, and should be folded into this subsection.

D Parameter Stability

A A cointegration test that fails should be diagnosed, not merely reported. This appendix sets out how instability is tested for in this chapter and what constraints govern reading a failure as evidence of regime change rather than as evidence of no long-run relation.

D.1 What the Shin statistic already tells us

A Shin's C statistic is Hansen's L_c restricted to intercept-only parameter variation, with $L_c^{\text{intercept}} = T \cdot C_{\text{Shin}}$. This identity holds to machine precision across the Specification A models.

B The consequence is interpretive and it matters. A Shin rejection is already evidence of parameter instability; it is not evidence of "no cointegration" in some undifferentiated sense. A report that observes a Shin rejection and stops there has not said what it appears to have said. It must either decompose the rejection with the full Hansen L_c, per parameter and jointly, or state plainly that the decomposition was not performed.

C One numerical caveat travels with this identity wherever both statistics appear: L_c and C_{Shin} will not agree numerically, because they use different long-run variance estimators — a score-process Bartlett estimator against the conditional $\omega_{u \cdot v}$ from the cointegrating regression. On this data they differ by roughly an order of magnitude. The discrepancy is expected and is not a sign that either is wrong.

D.2 Bootstrapped nulls, and why the direction of the null matters

A Test statistics introduced into this chapter take their inference from a bootstrapped null distribution rather than from hardcoded critical values, unless a properly indexed published table has been digitized in full.

B The construction of that null is where the single most dangerous error in this protocol lives. Tests whose null is cointegration with stable parameters — Hansen L_c, MeanF, SupF, and the CUSUM family — require resampling stationary residuals under a moving-block scheme. Tests whose null is no cointegration — Gregory-Hansen — require cumulating resampled differences into a random walk. Transposing the two data-generating processes silently inverts every p-value while producing output that looks entirely plausible. Block length follows the fitted long-run-variance bandwidth; i.i.d. resampling is not admissible, since it destroys exactly the serial dependence the block scheme exists to preserve.

D.3 Which specification the stability tests run on

A Endogenously selected step dummies are excluded from the stability tests. The CUSUM family and Gregory-Hansen run on the core specification — intercept plus stochastic regressors — because a dummy selected from the same data whose stability is being tested contaminates the test. Hansen L_c is reported both on the full fitted design and restricted to core parameters, and any table reporting it states which.

B The distinction between level shifts and vector shifts governs what may be claimed. Step dummies absorb level shifts only. They cannot absorb a shift in the cointegrating vector itself. A specification that answers detected breaks with step dummies has therefore not addressed slope instability, and is not described as break-robust anywhere in this chapter on that basis.

D.4 When a failure may be read as regime change

A Reading a cointegration failure as evidence of structural regime-switching is permitted only when four conditions hold jointly.

B Admissibility must not be the explanation: the integration-order gate passes for every regressor as actually constructed, lag structure included. Instability must be affirmatively demonstrated with bootstrapped p-values, not inferred from the failure itself. Break dates must be corroborated across the Hansen SupF argmax, the CUSUM supF argmax, Bai-Perron, and Gregory-Hansen — where these converge the reading is strengthened, and where they diverge the divergence is reported as divergence rather than resolved to whichever date suits the argument. And the constructive question must be answered: does a long-run relation appear once the cointegrating vector is allowed to break, by regime-split re-estimation or by Gregory-Hansen? If it does not, the honest verdict is that no stable long-run relation has been identified, and the text says so.

C Describing a cointegration failure as evidence of regime change on the strength of demonstrated instability alone, without the other three conditions, is not available.

D.5 Small samples

A Where a sub-sample falls to 30 observations or fewer, bootstrap p-values are reported alongside asymptotics. Below roughly 25 observations the estimate is flagged as underpowered and carries an explicit reliability caveat rather than a bare verdict. Step dummies that are constant within a sub-sample are dropped, and where this happens the text notes that the sub-sample specification therefore differs from the full-sample one.

B [**PLACEHOLDER**] Stability results are not yet reported here. Required: Hansen MATHPLACEHOLDER0XYZ (full design and core-restricted), MeanF, SupF, CUSUM, and Gregory-Hansen for each preferred specification, each with bootstrapped MATHPLACEHOLDER1XYZ-values under the correctly oriented null; the break-date corroboration table across the four detection methods; and the consolidated verdict summary. Source material is in the archived `reports/US_Section31_Stability_Audit/99_archive/`.

C

[**PLACEHOLDER**] One known defect to resolve before those results are reported: the residual ADF implementation used in the Chilean structural-break chain is not lag-augmented, and an unaugmented Dickey-Fuller regression on serially correlated residuals under-rejects severely. On the Specification A models the difference is decisive — 2 of 18 rejections at the 5% level with zero lagged differences against 16 of 18 under an information-criterion-selected lag order. Any prior reading of cointegration failure as regime-switching that rests on the unaugmented statistic must be revisited before it is repeated here.